



Diabetes & Other Disorders of the Endocrine Pancreas

Note: This reference document contains the content of the Canvas page for “Diabetes & Other Disorders of the Endocrine Pancreas” with minor modifications for presentation in a document format. Refer to this session’s Canvas page for the full interactive experience or if any hyperlinks do not function properly.

Table of Contents

Table of Contents	1
Overview	7
Learning Objectives	7
Preparation Content	7
Diabetes & Other Endocrine Pancreas Part 1 of 8: Diabetes Mellitus: Foundations and Frameworks 9	
Introduction	9
What Is Diabetes Mellitus?	9
Type 1 Diabetes Mellitus	9
Type 2 Diabetes Mellitus	10
Other Types of Diabetes	10
How Do Patients With Diabetes Mellitus Present?	10
Acute Presentation of Diabetes Mellitus	11
Acute Complications of Diabetes Mellitus	11
Chronic Complications of Diabetes Mellitus	12
Comorbidities of Diabetes Mellitus	13
What Is the Pathophysiology of Diabetes Mellitus?	13
How Do We Diagnose Diabetes Mellitus?	17
Screening for Chronic Complications	18
How Do We Manage Diabetes Mellitus?	18
Managing Type 1 Diabetes Mellitus	19
Management of Type 2 Diabetes Mellitus	19
Summary	20
Diabetes Mellitus: Foundations and Frameworks – Review Question	21
Diabetes & Other Disorders of the Endocrine Pancreas Part 2 of 8: Type 1 Diabetes Mellitus	22
Introduction	22



What Is Type 1 Diabetes Mellitus?	22
Epidemiology	22
How Do Patients With Type 1 Diabetes Mellitus Present?	23
Diabetic Ketoacidosis	23
What Is the Pathophysiology of Type 1 Diabetes Mellitus?	24
Immunology	24
Mechanism of Diabetic Ketoacidosis	25
Chronic Complications of Diabetes Mellitus	26
How Do We Diagnose Type 1 Diabetes Mellitus?	27
How Do We Manage Type 1 Diabetes Mellitus?	28
Glucose Control	28
Reduction of Complications	28
Monitoring Disease Activity and Complications	29
Summary	30
Type 1 Diabetes Mellitus – Review Question	31
Diabetes & Other Disorders of Endocrine Pancreas 3 of 8: Type 2 Diabetes Mellitus	32
What Is Type 2 Diabetes Mellitus?	32
Epidemiology	32
How Do Patients With Type 2 Diabetes Mellitus Present?	32
What Is the Pathophysiology of Type 2 Diabetes Mellitus?	33
Risk Factors	33
Disease Mechanism	34
Acute Complications	37
Chronic Complications	37
How Do We Diagnose Diabetes Mellitus Type 2?	38
Distinguishing Type 1 from Type 2 Diabetes Mellitus	40
How Can We Prevent and Manage Type 2 Diabetes Mellitus?	41
Prevention	41
Management: Lowering the Glucose	42
Monitoring and Preventing Diabetic Complications	43
Summary	44
Type 2 Diabetes Mellitus – Review Question	45
Diabetes & Other Disorders of the Endocrine Pancreas Part 4 of 8: Diabetic Ketoacidosis	46



Introduction	46
What Is Diabetic Ketoacidosis?	46
How Do Patients With Diabetic Ketoacidosis Present?	47
Urinary Symptoms	47
Cardiovascular Symptoms	47
Gastrointestinal Symptoms	47
Respiratory Symptoms	47
What Is the Pathophysiology of Diabetic Ketoacidosis?	48
Volume Depletion	50
Electrolyte Abnormalities	50
How Do We Diagnose Diabetic Ketoacidosis?	51
Laboratory Values	52
How Do We Manage Diabetic Ketoacidosis?	53
Volume Repletion	53
Insulin	53
Potassium and Phosphate	53
Chronic Management	55
Summary	56
Diabetic Ketoacidosis - Review Question	57
Diabetes & Other Disorders of the Endocrine Pancreas Part 5 of 8: Hyperosmolar Hyperglycemic State	58
Introduction	58
What Is Hyperosmolar Hyperglycemic State?	58
How Do Patients With Hyperosmolar Hyperglycemic State Present?	59
What Is the Pathophysiology of Hyperosmolar Hyperglycemic State?	59
Hyperglycemia and Volume Depletion	60
Potassium Abnormalities	61
Differentiating HHS from DKA	61
How Do We Diagnose Hyperosmolar Hyperglycemic State?	62
How Do We Manage Hyperosmolar Hyperglycemic State?	63
Normal Saline	64
Insulin	64
Potassium	64



Summary	65
Hyperosmolar Hyperglycemic State – Review Question	66
Diabetes & Other Disorders of the Endocrine Pancreas Part 6 of 8: Chronic Complications	67
Introduction	67
What Are the Chronic Complications of Diabetes Mellitus?	67
How Does Diabetes Mellitus Damage Tissues?	67
Hyperglycemic Injury in Diabetes Mellitus	68
Inflammation in Diabetes Mellitus	69
What Are the Macrovascular Complications of Diabetes Mellitus?	70
Coronary Artery Disease	70
What Is Diabetic Retinopathy?	71
Pathophysiology of Diabetic Retinopathy	71
Diagnosis of Diabetic Retinopathy	72
Management of Diabetic Retinopathy	72
What Is Diabetic Kidney Disease?	72
Clinical Presentation of Diabetic Kidney Disease	73
Pathophysiology of Diabetic Kidney Disease	73
Diagnosis of Diabetic Kidney Disease	74
Management of Diabetic Kidney Disease	75
What Is Diabetic Neuropathy?	75
Peripheral Neuropathy	75
Autonomic Neuropathy	77
Summary	79
Chronic Complications - Review Question	80
Diabetes & Other Disorders of the Endocrine Pancreas Part 7 of 8: Hypoglycemia	81
Introduction	81
What Is Hypoglycemia?	81
Fasting vs Reactive Hypoglycemia	81
How Do Patients With Hypoglycemia Present?	82
What Is the Pathophysiology of Fasting Hypoglycemia?	83
High Exogenous Insulin Levels in Patients with Diabetes Mellitus	83
High Endogenous Insulin Levels in Patients Without Diabetes Mellitus	84
Decreased Hepatic Gluconeogenesis and Glycogenolysis	84



Other Causes	86
What Is the Pathophysiology of Reactive Hypoglycemia?.....	86
Postoperative	86
How Do We Diagnose Hypoglycemia?	87
Confirmatory Tests	87
Testing for Chronic Medical Diseases	88
Tests for Insulinoma or Drug Ingestion	88
How Do We Manage Hypoglycemia?.....	89
Fasting Hypoglycemia	89
Reactive Hypoglycemia	89
Summary	90
Hypoglycemia - Review Question.....	91
<i>Diabetes & Other Disorders of the Endocrine Pancreas Part 8 of 8: Pancreatic Tumors</i>	<i>92</i>
Introduction	92
What Are Endocrine Pancreas Neoplasms?	92
What Are Insulinomas?	94
Clinical Presentation of Insulinomas	94
Pathophysiology of Insulinomas	94
Diagnosis and Treatment of Insulinomas	95
What Are VIPomas?	96
Clinical Presentation of VIPomas.....	96
Pathophysiology of VIPomas	96
Diagnosis and Treatment of VIPomas	96
What Are Glucagonomas?	97
Clinical Presentation of Glucagonomas	97
Pathophysiology of Glucagonomas.....	98
Diagnosis and Treatment of Glucagonomas	98
What Are Somatostatinomas?	99
Clinical Presentation of Somatostatinomas	99
Pathophysiology of Somatostatinomas	99
Diagnosis and Treatment of Somatostatinomas	99
Summary	100
Pancreatic Tumors - Review Question	101



<i>Diabetes & Other Disorders of the Endocrine Pancreas- Review Questions and Answers</i>	<i>102</i>
Diabetes Mellitus: Foundations and Frameworks - Review Question	102
Type 1 Diabetes Mellitus - Review Question	103
Type 2 Diabetes Mellitus - Review Question	103
Diabetic Ketoacidosis - Review Question	104
Part Hyperosmolar Hyperglycemic State - Review Question	104
Chronic Complications - Review Question	105
Hypoglycemia – Review Question	105
Pancreatic Tumor – Review Question	106



Overview

This session is created to integrate all the basic sciences concepts regarding diabetes. This session will require you to use your prior knowledge of the pancreas and its functioning. You could easily relate to the case-based discussion using earlier discussions on this topic. During this session, you will be working through a few cases which will help you integrate Pathology, Pharmacology, and Clinical Medicine.

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Learning Objectives

By the end of this session, you will be able to meet the following learning objectives:

1. Differentiate between Type 1 and Type 2 Diabetes Mellitus, including:
2. Etiology and etiologic mechanisms
 - Pathophysiology
 - Morphological features
 - Clinical presentation and complications (acute and chronic)
 - Diagnostic investigations and screening criteria
 - Principles of treatment and long-term management
3. Explain hyperglycemic emergencies, including Diabetic Ketoacidosis (DKA) and Hyperosmolar Hyperglycemic State (HHS), focusing on:
 - Pathophysiological basis
 - Clinical features and differentiation
 - Management strategies
4. Describe and discuss the complications of diabetes, both acute and chronic, with attention to:
 - Pathogenesis and morphological changes in target organs
 - Diagnostic approaches and clinical implications
 - Prevention strategies and current treatment options
5. Identify and discuss other forms of diabetes, including:
 - Secondary diabetes (e.g., drug-induced, endocrinopathies, pancreatic disease)
 - Genetic and syndromic forms (e.g., MODY, neonatal diabetes)
6. Discuss selected pancreatic endocrine tumors, including insulinoma and Zollinger-Ellison syndrome, under the headings of:
 - Etiology and pathophysiology
 - Morphology and diagnostic work-up
 - Clinical features and management principles

Preparation Content



For this session, content has been sectioned into separate sections. Visit these sections below to review content for each topic.

Diabetes & Other Endocrine Pancreas Part 1 of 8: Diabetes Mellitus: Foundations and Frameworks

Introduction

Case Connection

MC is a healthy 13-year-old male who has been playing soccer for several years. His parents are concerned that he asked to be taken out of a game last weekend because he felt tired and dizzy. His coach told them that MC also sat out of practices this week. His appetite seems fine, and he drinks a lot of water. "He's constantly in the bathroom," his mom says. His weight is down 10 lb since his sports physical 6 months ago, but his exam is otherwise normal.

How will you explain the diagnosis to MC and his parents? Consider your answer as you read, and we'll revisit MC at the end of the session.

What Is Diabetes Mellitus?

Diabetes mellitus (DM) is a group of metabolic disorders characterized by high blood glucose. Blood glucose is proof that you can have too much of a good thing. Although glucose serves a critical role as fuel for many of our bodily functions, it must remain in a very tightly controlled range. If the level goes too low, you can fall into a coma. If glucose is too high, it can damage tissues throughout the body. When a person's blood glucose is consistently too high and the mechanisms that return it to the normal range fail, the person has DM. Nearly 10% of the US population has a form of DM, so it is critical that clinicians understand the disease and be prepared to care for patients with the diagnosis.

DM has two major types, 1 and 2.

Type 1 Diabetes Mellitus

In type 1 DM, the pancreatic islet cells undergo autoimmune destruction, resulting in **inadequate** or **no insulin release** to regulate the patient's glucose levels. This form presents most often in children.

Clinical Correlation

Insulin is a hormone released by the pancreatic β cells in the islets of Langerhans, which make up the endocrine portion of the pancreas. The other part, the exocrine pancreas, is responsible for some aspects of digestion, as it secretes enzymes into the duodenum to help break down carbohydrates, proteins, and lipids. But the hormone-secreting endocrine pancreas is what is involved in DM.

Type 2 Diabetes Mellitus

In type 2 DM, the body's response to insulin is decreased. This is referred to as **insulin resistance**. The body's cells are not properly able to respond when insulin is released to lower the blood glucose level. In addition, the endocrine pancreas becomes deficient and cannot raise the insulin levels to compensate for the insulin resistance. This leads to an elevated level of glucose in the blood. Type 2 DM presents most often in obese adults, but obese children and nonobese patients can get it, too. It also has a strong hereditary component.

Knowledge Check

Question: What is insulin resistance?

Answer: Insulin resistance is the body's decreased response to insulin.

Other Types of Diabetes

Occasionally, patients have a form of diabetes that is not type 1 or type 2. Maturity onset diabetes of the young (MODY) is a group of rare disorders that typically cause mild-to-moderate glucose intolerance resulting from mutations in genes responsible for insulin secretion by pancreatic β cells. Consider the diagnosis of MODY in patients with autosomal dominant inheritance, early onset of hyperglycemia (usually <25 years; sometimes in the neonatal period), and impaired insulin secretion.

How Do Patients With Diabetes Mellitus Present?

Although types 1 and 2 DM present at different ages, the diagnostic criteria are the same. For both types of DM, the symptoms of uncontrolled disease are similar, as are the main chronic complications. We'll now compare and contrast the acute and chronic complications.



Acute Presentation of Diabetes Mellitus

Whether from type 1 or type 2 DM, severe hyperglycemia presents with polyuria, polydipsia, and polyphagia. The excess glucose in the bloodstream causes hyperglycemia. The extra glucose is then filtered into the kidney tubules, drawing water along with it and causing polyuria in a process called **osmotic diuresis**. The patient then gets abnormally thirsty (polydipsia). Because glucose cannot enter cells, a feeling of low energy and increased appetite (polyphagia) can result, especially in type 1 DM. Note that type 1 DM typically presents with weight loss, whereas type 2 DM is associated with weight gain in most cases (because weight gain leads to insulin resistance).

Another difference is that although type 1 DM commonly presents with acute illness and severe symptoms in a previously well child, type 2 DM is often a slower, asymptomatic progression that gets diagnosed by the patient's primary care provider, who watches the at-risk patient's glucose levels gradually rise. This means that type 2 DM is more likely to be treated before any of the acute symptoms occur.

Knowledge Check

Question: Remember the three Ps of DM: Polyphagia, Polyuria, and Polydipsia

Answer: Three Ps of DM

Acute Complications of Diabetes Mellitus

Besides the acute hyperglycemic presentation described above, patients with types 1 or 2 DM are at risk for acute complications that can occur when they have other major illnesses, when they stop their medicine, and when major body stressors are present. The stress leads to severe hyperglycemia, which, in turn, can lead to **diabetic ketoacidosis (DKA)** or **hyperosmolar hyperglycemic state (HHS)**. Both conditions can be fatal if untreated. Note that some patients with DM first present for medical care with one of these acute illnesses.

Diabetic Ketoacidosis. DKA is mostly seen in patients with type 1 DM and is caused by the lack of insulin combined with stress. The lack of insulin leads to increased lipolysis (conversion of triglycerides to free fatty acids), which are metabolized in the liver to ketoacids. This process leads to hyperglycemia, high serum ketones, and metabolic acidosis.

Hyperosmolar Hyperglycemic State. HHS, in contrast, is more common in patients with type 2 DM and is characterized by extremely high blood sugar levels without ketosis, because insulin is still present. The key clinical features that distinguish DKA from HHS are summarized in Table 1.



Table 1

	DKA	HHS
Onset	Hours to days	Days to weeks
Respirations	Deep (Kussmaul) respirations	Usually normal
Neurologic symptoms	Less common	More common
Abdominal pain	More common	Less common
Volume status	Low	Low
Urine and serum ketones	Present	Absent
Serum pH	Low	Normal

Knowledge Check

Question: What are potentially fatal acute complications of types 1 and 2 DM?

Answer: DKA is mostly seen in patients with type 1 DM, whereas HHS is more common in patients with type 2 DM.

Chronic Complications of Diabetes Mellitus

Chronic hyperglycemia (over 5 years) may cause severe chronic complications. The goal for management of both types of DM is to avoid such long-term chronic complications. Most of these complications result from **nonenzymatic glycosylation**, in which glucose attaches to proteins without the help of an enzyme. Nonenzymatic glycosylation leads to inflammation and vascular damage in many different tissues, including the retina, kidney, and nerves. Hyperglycemia also leads to the irreversible glycosylation of the hemoglobin in red blood cells (RBCs) to form **hemoglobin A_{1c}**, an important diagnostic test for DM that is used to assess the patient's average glucose levels over a 3-month period.



Knowledge Check

Question: Mutations of which human leukocyte antigens are associated with the development of type 1 DM?

Answer: Mutations in HLA-DR3 and HLA-DR4 are associated with the development of type 1 DM.

Chronic complications of DM include:

- Microvascular complications, including diabetic retinopathy (causing blindness) and diabetic kidney disease (causing proteinuria and nephropathy)
- Cardiovascular complications resulting from atherosclerosis, including angina, myocardial infarction, heart failure, and stroke
- Neurologic complications including peripheral neuropathy (numbness, burning, pain) and autonomic nerve disease (orthostatic hypotension)
- Gastrointestinal complications (nausea/vomiting) resulting from loss of neural input and control of stomach function, termed diabetic gastroparesis

The most effective way to prevent most of these complications is to establish excellent control of the serum glucose early in the disease course and maintain it long term.

Clinical Correlation

DM is a risk factor for many diseases, including dementia, infections, erectile dysfunction, female infertility, and periodontal (gum) disease.

Comorbidities of Diabetes Mellitus

Type 2 DM is often preceded or accompanied by **metabolic syndrome**, characterized by abdominal obesity, hypertension, low levels of high-density lipoproteins, hypertriglyceridemia, and (in nondiabetic patients) elevated serum glucose levels.

Changes in lipid metabolism lead to higher incidences of nonalcoholic fatty liver disease, whereas increased fat around the neck can cause obstructive sleep apnea. Recent studies have also found links between DM and some forms of cancer.

What Is the Pathophysiology of Diabetes Mellitus?

Both types of DM have genetic components. Type 1 DM is associated with mutations in human leukocyte antigens HLA-DR3 and HLA-DR4, but the mutations are only weakly linked with inheritance. Type 2 DM has a stronger genetic component, and its identical twin concordance rate is 70% to 90%.



Knowledge Check

Question: Mutations of which human leukocyte antigens are associated with the development of type 1 DM?

Answer: Mutations in HLA-DR3 and HLA-DR4 are associated with the development of type 1 DM.

Although both types are characterized by hyperglycemia, each disease reaches this state by very different mechanisms. Type 1 DM results from an **autoimmune destruction of the pancreatic islet cells**, a type of immune hypersensitivity reaction. The destruction is both by antibodies (autoantibodies against islet cells) and by cell-mediated immunity, in which CD8⁺ lymphocytes invade the islets, specifically targeting and destroying the β cells.

In contrast, type 2 DM is caused by a **mix of insulin resistance and inadequate insulin secretion from islet cells**. Insulin resistance is correlated with obesity, sedentary lifestyle, high triglycerides, inflammation, and secretion of certain cytokines by adipocytes (adipokines), but the cause of insulin resistance is still unclear (Figure 1).

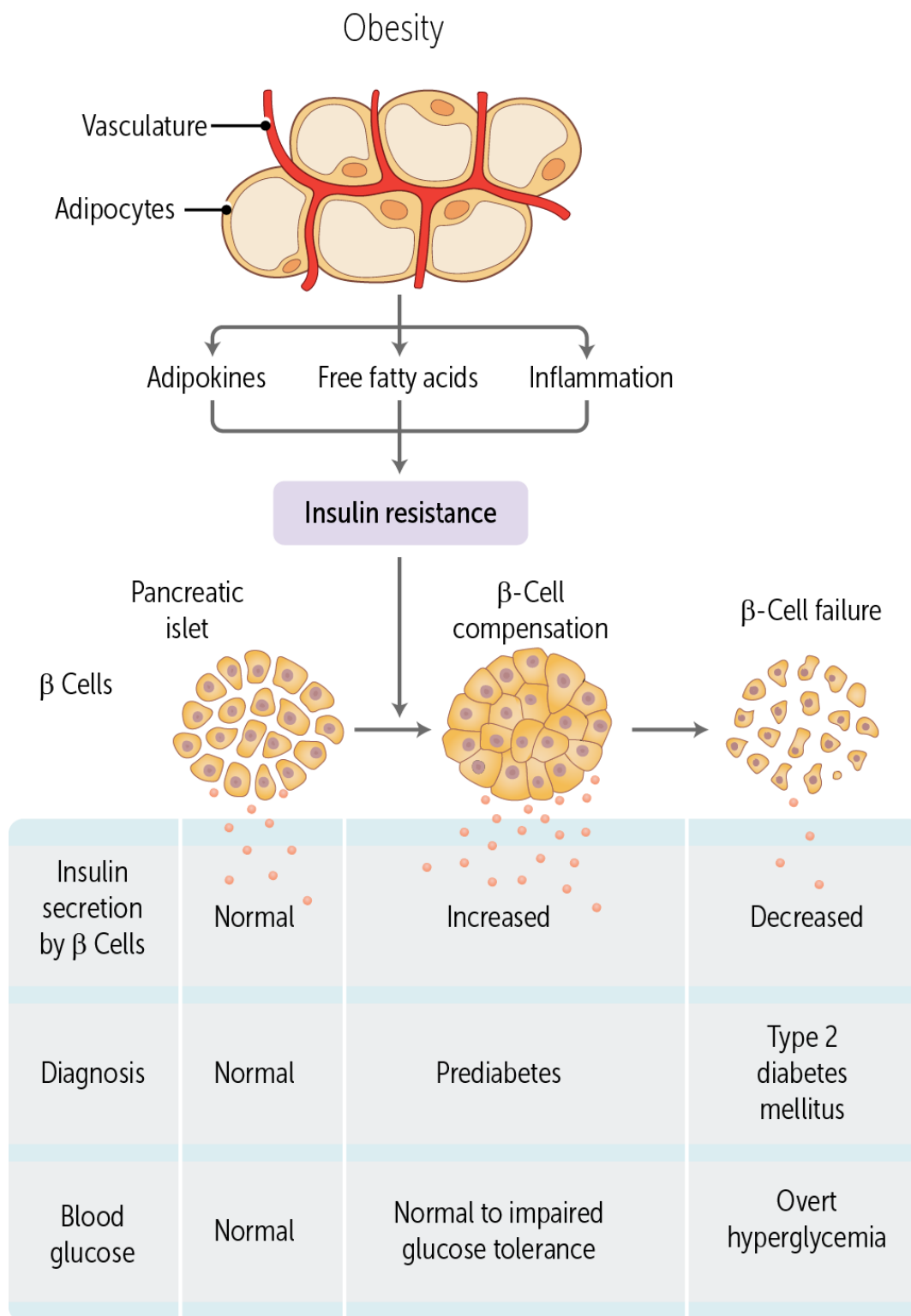


Figure 1. Image Credit: ©ScholarRx, used with permission.

Insulin resistance leads to β -cell hypertrophy with mildly elevated blood glucose levels and increased insulin secretion. This is termed **prediabetes**, because patients do not yet



meet diagnostic criteria for type 2 DM. Eventually their serum glucose may rise into diabetic range (type 2 DM), and after that, β -cell failure occurs. At this point, some patients with type 2 DM require insulin therapy, just like patients with type 1 DM.

Another significant difference between the two types is body weight, especially body fat. Because type 2 DM has a strong correlation with metabolic syndrome, patients are often overweight or obese, whereas patients with type 1 DM are usually thin. Visceral (abdominal) fat seems to be particularly important in type 2 DM, perhaps more than the patient's overall weight.

Clinical Correlation

Metabolic syndrome is defined as having at least three of the following symptoms: abdominal obesity, hypertension, elevated glucose levels, low levels of high-density lipoproteins, and hypertriglyceridemia. Patients with these symptoms are at higher risk for type 2 DM and for cardiovascular disease.

Table 2 summarizes key differences between type 1 and type 2 diabetes.

Table 2

	Type 1	Type 2
Primary defect	Immune destruction of β cells	$\uparrow$ Insulin resistance
Insulin treatment	Always	Only when oral agents and diet fail
Association with obesity	No	Usually
Genetic predisposition	Weak, polygenic	Strong, polygenic
Association with HLA system	HLA-DR3 and HLA-DR4	No
Glucose intolerance	Severe	Mild to moderate
Ketoacidosis	Common	Rare
Pancreatic β cells	Destroyed by an autoimmune reaction	Decreased insulin secretion but present



	Type 1	Type 2
Typical serum insulin levels	Low or zero	Variable, often high
Patients with autoantibodies against islet cells	85%-95%	5%-10%

How Do We Diagnose Diabetes Mellitus?

In addition to noting the characteristic symptoms, clinicians use various glucose tests to make a diagnosis of either type of DM. An abnormal value in any of the four tests listed in Table 3 is a strong indicator of the presence of DM.

Table 3

Test	Cutoff value	Notes
Fasting blood glucose	≥ 126 mg/dL	Requires fasting for ≥ 8 hours
Glycated hemoglobin (HbA _{1c})	$\geq 6.5\%$	The test results may be influenced by certain conditions (eg, hemolytic anemia and end-stage renal disease)
Random venous blood glucose	≥ 200 mg/dL	Used only for patients exhibiting symptoms of hyperglycemia
Oral glucose tolerance test	Serum glucose ≥ 200 mg/dL 2 h after the patient ingests large load of glucose	Cumbersome, and used less now

In the absence of unequivocal hyperglycemia, DM is diagnosed by two abnormal test results. Note that the random blood glucose is only valid in patients with symptoms, while the other tests can be used for either symptomatic or asymptomatic patients.

HbA_{1c} levels are used to track glucose levels over an extended period. Because the glycated hemoglobin remains in the body until RBC turnover occurs (every 120 days), HbA_{1c} is useful for tracking how well patients regulate their glucose levels over time. It

may also be used for initial diagnosis. Figure 2 shows how the levels change with increasing serum glucose concentrations.

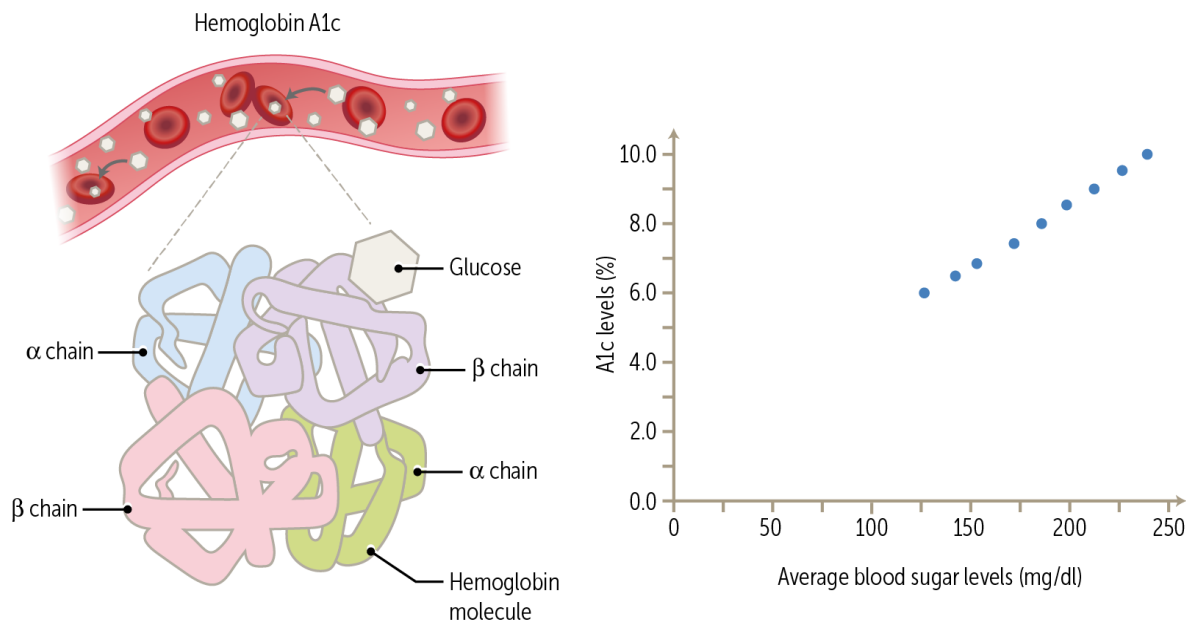


Figure 2. Image Credit: ©ScholarRx, used with permission.

Screening for Chronic Complications

Many chronic complications are preventable with early detection and treatment. For example, annual tests of urine albumin levels can be used to detect nephropathy in diabetic kidney disease, and annual eye exams by an ophthalmologist can help detect retinopathy. In addition, foot exams and neurologic tests help gauge the degree of neurologic and circulatory damage caused by hyperglycemia.

Knowledge Check

Question: What is one way to screen for kidney disease in a patient with DM?

Answer: Annual tests of urine albumin levels

How Do We Manage Diabetes Mellitus?

Patients with either type of DM benefit from a balanced diet high in fiber and low in both fat and refined, simple carbohydrates like sucrose and fructose. The highly restrictive low-carb diets of the past are no longer recommended for most patients.



Patients with DM also require careful control of other comorbidities that may cause cardiac disease. Smoking cessation, hypertension control, and use of statins to lower serum lipids are routine parts of diabetic care.

Managing Type 1 Diabetes Mellitus

Type 1 DM is treated with **insulin replacement therapy** because patients have little to no endogenous insulin production. Many different preparations of insulin are available, including long-acting, intermediate, and short-acting types. All run the risk of hypoglycemia, and therapy must be followed by multiple daily finger-stick glucose checks or with a real-time continuous glucose monitoring system. Alternatively, insulin pumps can be installed to inject insulin automatically but still require users to monitor their glucose levels.

Management of Type 2 Diabetes Mellitus

Type 2 DM is typically managed first by exercise and weight loss. If these steps are not effective, patients take oral antihyperglycemic medications to lower their glucose levels. These drugs work through a variety of mechanisms, including increasing insulin release, increasing excretion of urinary glucose, and decreasing insulin resistance. Patients with long-standing type 2 DM may eventually require insulin therapy as their endogenous insulin secretion drops.

Knowledge Check

Question: How does initial treatment differ between type 1 DM and type 2 DM?

Answer: Patients with type 1 DM are started on insulin because they produce little or no endogenous insulin; patients with type 2 DM are first told to exercise and lose weight, then are started on antihyperglycemic medications if weight loss does not solve the problem.



Case Connection

Thinking back to MC, how do you explain the diagnosis to him and his parents? MC's serum glucose is greatly elevated, and he has glucose in his urine. You explain to MC and his parents that he has type 1 DM. You describe the basics of dietary modifications, blood sugar monitoring, and insulin therapy. MC's father says, "My boss has diabetes, and he just takes a pill." You explain that with type 1 DM, insulin is the best treatment because in this type, the pancreas does not secrete insulin. (You also explain that in type 2 diabetes, insulin is secreted but there is resistance to it, so pills are usually more appropriate for type 2.) You arrange admission to the hospital for observation, where MC will start insulin therapy and receive more intensive education about type 1 diabetes. You also make an appointment for MC to return to the clinic in a few days.

Summary

- Both types of diabetes mellitus (DM) present with hyperglycemia, polydipsia, polyuria, and polyphagia.
- Type 1 DM usually presents at younger ages; it is associated with a thin or normal body weight and mutations in HLA-DR3 or HLA-DR4.
- Type 2 DM typically presents in adulthood, has a strong genetic component, and is associated with metabolic syndrome.
- Chronic complications of both types of DM include retinopathy, diabetic kidney disease, neuropathy, hyperlipidemia, infections, and cardiovascular disease.
- Severe complications of DM include diabetic ketoacidosis (more common with type 1 DM) and hyperosmolar hyperglycemic state (more common with type 2 DM).
- DM is characterized by hyperglycemia resulting from inadequate control of blood glucose by insulin.
- Type 1 DM is caused by a lack of insulin, arising from the autoimmune destruction of the pancreatic islet cells.
- Type 2 DM is caused by a combination of insulin resistance and inadequate insulin secretion.
- DM is most often diagnosed with a fasting blood glucose of at least 126 mg/dL or a glycated hemoglobin (HbA1c) level of at least 6.5%.
- Many chronic complications are preventable with early detection and treatment, so regular screening is recommended.
- Treatment for type 1 DM is with insulin.
- Treatment for type 2 DM usually includes weight loss and oral antihyperglycemic drugs, with insulin used later in the course of the disease.



Diabetes Mellitus: Foundations and Frameworks – Review Question

Which of the following tests provides the most accurate measurement of glycemic control over time?

- Fasting glucose
- Finger-stick glucose
- Glucose on urine dipstick
- Hemoglobin A_{1c}
- Oral glucose tolerance test

Diabetes & Other Disorders of the Endocrine Pancreas Part 2 of 8: Type 1 Diabetes Mellitus

Introduction

Case Connection

You are working on the medical staff for a summer camp. In the early morning, DA, a 13-year-old boy with asthma, comes to you breathing very rapidly and reporting that he doesn't feel well. He appears to be dehydrated, but you recall one of the other campers saying that DA always drinks more than his share of their water rations. DA has used his inhaler several times this morning, but he isn't improving. "I think it's broken," he says. "Can you ask my mom to bring me a new one?" Knowing that diabetic ketoacidosis can present with rapid breathing, you quickly arrange for transport to the nearest hospital. While you wait, you convince DA to let you check his blood sugar, and his glucose is greatly elevated. You need to call DA's parents, who are out of state.

What will you tell DA and his parents about his condition? Consider your answer as you read, and we'll revisit DA at the end of the brick.

What Is Type 1 Diabetes Mellitus?

Diabetes mellitus (DM) is a group of metabolic disorders characterized by high blood glucose. Here, we'll address type 1 DM, formerly known as juvenile diabetes. Like the more common type 2 DM, it leads to severe acute and chronic complications like heart, kidney, and eye disease. Unlike type 2 DM, type 1 DM is caused purely by a lack of insulin secretion by the endocrine pancreas. (Type 2 is initially caused by peripheral resistance to insulin.) When insulin is deficient or absent, high blood sugar levels (hyperglycemia) can occur, leading to many of the adverse effects of the disease.

Epidemiology

Type 1 DM accounts for 5% to 10% of all diabetes cases worldwide and for 60% of diabetes diagnoses in children and adolescents in the United States. Distribution of age at onset is bimodal, with most cases diagnosed from 4 to 6 or 10 to 14 years old. However, incidence is varied on the basis of multiple factors such as geography, age, sex, family history, and ethnicity. The incidence is higher among males than females.



While there has been much discussion about rising rates of type 2 diabetes mellitus in children, largely because of obesity, the incidence of type 1 diabetes has also risen in the United States, going from 20 cases per 100,000 in 2002 to 23 cases per 100,000. The reasons for this are unknown.

Knowledge Check

Question: Which kind of diabetes is most commonly diagnosed in children?

Answer: Type 1 DM is most commonly diagnosed in children.

How Do Patients With Type 1 Diabetes Mellitus Present?

Patients with type 1 DM usually present at age 18 years or younger. They normally present with symptoms, sometimes severe. This contrasts with the often asymptomatic presentation of type 2 DM. The patient may describe experiencing symptoms of high blood glucose, which are the three Ps: **polyphagia** (increased hunger), **polyuria** (high volumes of dilute urine), and **polydipsia** (increased thirst). Polyuria occurs because any serum glucose value of more than 240 mg/dL will overwhelm the resorptive capacity of the renal proximal tubules and lead to glucose spilling into the urine. This draws water into the tubule and increases urine volume. Polydipsia is the patient's attempt to keep up with the water loss by drinking lots of fluids. Patients often experience weight loss because of low cellular glucose uptake; this also causes feelings of weakness and high appetite (polyphagia). These symptoms may prompt a clinic visit, leading to the diagnosis.

Knowledge Check

*Question: Remember the **three Ps** of DM: **Polyphagia**, **Polyuria**, and **Polydipsia***

Answer: Three Ps of DM

Diabetic Ketoacidosis

Some patients initially present with the more severe syndrome of **diabetic ketoacidosis (DKA)**, a severe, life-threatening combination of hyperglycemia and metabolic acidosis. This can also occur after diagnosis if the patient stops taking insulin or if an infection or other severe disease causes the type of stress that increases insulin requirements. In DKA, patients may present with deep labored respirations (Kussmaul respirations), abdominal pain, hypotension, and signs of severe dehydration. If untreated with insulin and fluids, DKA can be fatal.

What Is the Pathophysiology of Type 1 Diabetes Mellitus?

The specific causes of type 1 DM remain unclear but likely involve both genetic and environmental factors. The lifetime risk of developing type 1 DM is higher in patients with close relatives who have the condition, and the genetic predisposition has been shown by twin concordance studies. The overall mechanism is T and B cell-mediated immune destruction of the **pancreatic β cells** in the islet of Langerhans. These cells are responsible for insulin secretion, so their destruction leads to absent or low insulin, with consequent hyperglycemia.

Immunology

Type 1 DM is an autoimmune disorder that is thought to be precipitated by environmental factors in a genetically susceptible individual. Initially, **autoantigens** arise within pancreatic β cells and are subsequently targeted by the immune system, possibly as the result of molecular mimicry after a viral infection. If a viral protein is antigenically similar to a β -cell molecule, cross-reactivity may develop, leading to an autoreactive immune response that destroys the β cell. Major histocompatibility complex haplotypes HLA-DR3 and HLA-DR4 are most strongly associated with type 1 DM.

Immunologically, type 1 DM is an example of a **type IV cell-mediated hypersensitivity reaction**. Antigen-presenting cells present β -cell autoantigens to helper T cells, which release various cytokines in response. These cytokines act to recruit T and B lymphocytes, which cause direct cytotoxic and autoantibody-mediated destruction of pancreatic β cells. This destruction is initially seen as **lymphocytic infiltration or insulinitis** in a histological section; over time, there is a gradual loss of β cells in the islets (Figure 1).

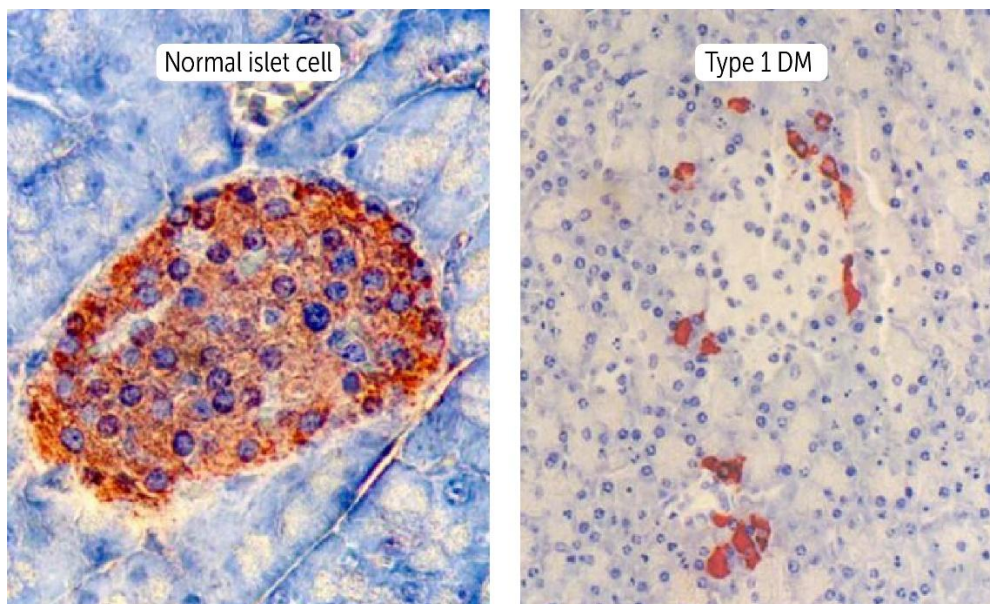


Figure 1. Image credit: © Tabiin et al; licensee BioMed Central Ltd. 2004 J Autoimmune Dis. 2004;1:3.

These autoantibodies target both pancreatic islet β cells and glutamic acid decarboxylase, an enzyme that has a role in controlling insulin release from β cells. This culminates in the absolute destruction of pancreatic β cells. Patients with type 1 DM typically have hyperglycemia once β -cell loss exceeds 80%.

Knowledge Check

Question: What pathophysiological mechanism leads to the destruction of pancreatic β cells?

Answer: An autoimmune mechanism leads to the destruction of pancreatic β cells, as they are destroyed by autoantibodies and cellular immunity.

Mechanism of Diabetic Ketoacidosis

As mentioned, many patients with type 1 DM present with DKA, showing hyperglycemia, high blood ketones, and metabolic acidosis. The combination of low-insulin and high-insulin counterregulatory hormones leads to the production of the **ketone bodies**, which are the root causes of DKA (Figure 2).

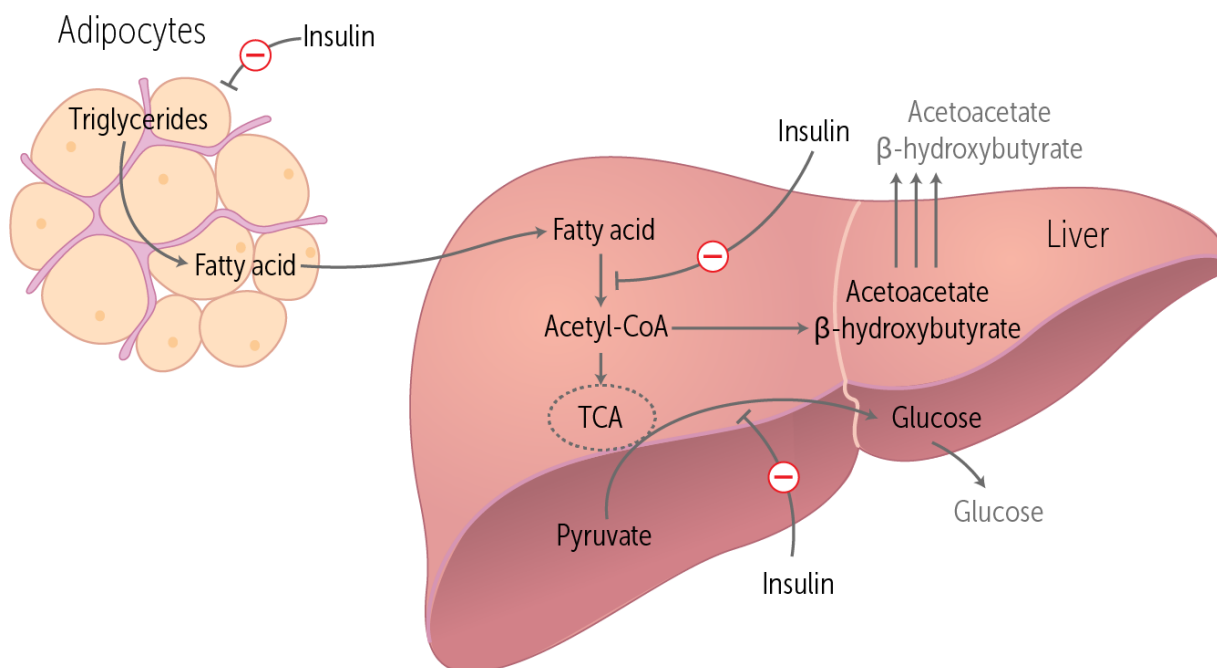


Figure 2. Image Credit: ScholarRx, used with permission.

In DKA, because there is no insulin to drive glucose into the cells, the body breaks down fat to use as fuel. Lipolysis converts triglycerides (TGs) to acetyl coenzyme A, normally a substrate for the TCA cycle. In DKA excess acetyl CoA is then converted to acidic ketone

bodies (ketoacids) in the liver. These enter the blood and cause systemic **metabolic acidosis**.

In DKA, severe hyperglycemia causes **osmotic diuresis** because the kidneys reabsorb only some of the excess glucose that is filtered. The high glucose in the urine pulls water and sodium into the urine, leading to high urine volume (polyuria) and dehydration.

Knowledge Check

Question: Why are ketones produced in patients with type 1 DM?

Answer: Ketones are produced as an energy source in patients with type 1 DM because the lack of insulin prevents cellular uptake and use of circulating glucose.

Chronic Complications of Diabetes Mellitus

Patients with either type of DM may develop one or more chronic complications. These are usually more prevalent and severe if glucose control has been poor and are often due to the **nonenzymatic glycation of proteins** that occurs when serum glucose levels are chronically high. These glycated proteins are proinflammatory and lead to both microangiopathy and macroangiopathy, resulting in thickening of the vessel walls and reduced blood supply to the tissues. These common chronic complications include:

- Retinopathy, which can lead to blindness
- Diabetic kidney disease, which can lead to chronic kidney disease and the need for dialysis or transplant
- Peripheral neuropathy, which can cause numbness, pain, loss of sensation, and, when combined with trauma, ulceration in the feet
- Autonomic neuropathy, which can cause orthostatic hypotension, gastrointestinal dysfunction (nausea, vomiting, diarrhea), and urinary retention or incontinence
- Accelerated atherosclerosis, with higher risks of coronary artery disease, stroke, and peripheral artery disease



How Do We Diagnose Type 1 Diabetes Mellitus?

A diagnosis of DM (both types 1 and 2) is typically made with one or more blood glucose tests (Table 1).

Table 1

Test	Cutoff value	Description
Fasting blood glucose	≥ 126 mg/dL	Requires two positive test results
Glycated hemoglobin (HbA _{1c})	$\geq 6.5\%$	Used more frequently with patients with type 2 DM; value $< 6.5\%$ does not exclude the diagnosis of type 1 DM
Random venous blood glucose	≥ 200 mg/dL	Used only for patients exhibiting symptoms of hyperglycemia
Oral glucose tolerance test	Serum glucose ≥ 200 mg/dL 2 h after the patient ingests glucose	Not frequently used now

The glycated hemoglobin (HbA_{1c}) test measures the percentage of glycated red blood cells. HbA_{1c} levels increase as glucose levels increase and show how well controlled the patient's blood sugar levels have been over the past 3 months. (The life cycle of red blood cells is 120 days, or approximately 4 months.) Although the test is still controversial for the initial diagnosis of type 1 DM, it is quite useful for monitoring long-term glucose control.

If a patient has an abnormal value in any of the four tests shown in the table, the diagnosis of DM should be high on the differential and correlated clinically. In the absence of unequivocal hyperglycemia, results should be confirmed by repeat testing.

Knowledge Check

Question: What does an elevated hemoglobin A1c test signify?

Answer: Elevated hemoglobin A1c signifies persistent hyperglycemia, leading to nonenzymatic glycosylation of red blood cell membranes.



How Do We Manage Type 1 Diabetes Mellitus?

Several principles must be included in comprehensive management of type 1 DM. However, since this is a disease of insulin shortage or absence, all patients require **insulin therapy**. Treatment can be divided into glucose control and reduction of vascular complications.

Glucose Control

To attain glucose control, the diet should be high in complex carbohydrates and fiber but low in fat and refined sugars such as fructose and sucrose. Insulin is typically injected multiple times per day, combining a long-acting agent with short-acting insulin taken at meals. Insulin pumps may be used in patients who dislike administering multiple daily insulin shots.

Pancreatic transplants are done along with renal transplants in patients who have end-stage kidney disease and require renal transplantation. Sometimes, pancreatic transplants are also done in patients without renal disease who have severe diabetic complications. Islet cell transplants are under investigation.

Reduction of Complications

All patients should be routinely monitored for chronic complications of DM. This includes:

- **Hypertension:** Blood pressure should be <140/90 mm Hg and <130/80 mm Hg if cardiovascular risk is increased. An angiotensin-converting enzyme (ACE) inhibitor is the best drug for initial hypertension treatment.
- **Hyperlipidemia:** Statin therapy should be used for all patients with DM and cardiovascular disease or high cardiac risk factors. The American Diabetes Association also recommends statin therapy for diabetic patients aged 40 to 75 years even without these risk factors or hyperlipidemia.
- **Retinopathy:** Annual ophthalmology exams are recommended.
- **Nephropathy:** Annual determination of urine albumin, an early sign of diabetic kidney disease, should be taken. An ACE inhibitor should be started if urine albumin exceeds 30 mg/24 hours or 30 mg albumin per gram creatinine on a spot urine specimen.
- **Neuropathy:** Annual in-office exams of the feet should be done for ulcers and for sensory loss; sensory loss should be determined with pinprick, vibratory, and microfilament testing.

Drugs for Reducing Complications. Since both hypertension and hyperlipidemia are strong risk factors for diabetic complications, they should be aggressively managed.



- **Blood pressure control** should target a blood pressure under 140/90 mm Hg, or 130/80 mm Hg if there is concurrent cardiovascular disease. Multiple drugs may be needed. An ACE inhibitor like enalapril is often used to do this, given its beneficial effects on the heart and kidney. ACE inhibitors should always be used for diabetic kidney disease. This is because these drugs delay the progression, in part because they reduce glomerular capillary pressure by dilating the efferent arteriole.
- **Statin therapy** is used to treat hyperlipidemia and is needed in all adult patients with DM who have high cardiovascular risk factors or who are at least 40 years of age.

Knowledge Check

Question: What drug would benefit a patient with type 1 DM and hypertension?

Answer: Use of ACE inhibitors would benefit a patient with type 1 DM and hypertension because these drugs have beneficial effects on the heart and kidney, especially in patients with diabetic kidney disease.

Monitoring Disease Activity and Complications

Close monitoring of blood glucose levels is essential, both to prevent the complications seen with hyperglycemia and the potentially lethal complication of hypoglycemia. **Home monitoring of blood glucose** levels should be done throughout the day (4-5 times daily) to avoid both hypoglycemia and hyperglycemia. This may be accomplished with fingerstick monitors or the newer continuous glucose monitors ("artificial pancreas"). Patients should be asked about hypoglycemic symptoms (tachycardia, dizziness, sweating) at each office visit.

Hemoglobin A_{1c} levels should be checked every 3 months. Treated patients should have a **value $\leq 6.5\%$** to indicate good control. Trying to reach lower values risks hypoglycemia, so the target value is usually relaxed to $<8\%$ in older patients in whom hypoglycemic falls may be problematic.



Case Connection

Thinking back to DA, what do you tell him and his parents about his condition? You explain to DA that he likely has type 1 diabetes mellitus, which is caused by his pancreas failing to make the insulin needed to keep his blood sugar in the normal range. "I haven't had any sugar today," DA says. "Right, but if your body stops making insulin, the sugars just stay in the blood for a long time, and that makes you sick," you say. You encourage his parents to come to the hospital because DA will likely be admitted and given intravenous fluids and insulin. You assure them over the phone that you will stay with him until they arrive.

Summary

- Type 1 diabetes mellitus (DM) is characterized by insufficient insulin production.
- Type 1 DM most commonly presents in childhood.
- Classical symptoms of high glucose include polyphagia, polyuria, and polydipsia.
- Compared to patients with type 2 DM, patients with type 1 DM are more likely to present with diabetic ketoacidosis on presentation or as a complication.
- Type 1 DM is primarily caused by autoimmune destruction of the insulin-producing β cells of the pancreas.
- Autoantibodies target both pancreatic β cells and glutamic acid decarboxylase, an enzyme that has a role in controlling insulin release from β cells.
- Fasting blood glucose tests, oral glucose tolerance tests, random blood glucose tests, and glycated hemoglobin (HbA1c) tests are used to diagnose and manage diabetes mellitus.
- Glycated hemoglobin can be used not only for diagnosis, but also to monitor the long-term control of blood sugar.
- Management of type 1 DM includes a balanced diet, insulin injections, consistent home monitoring of glucose levels, and good control of blood pressure and serum lipids.
- The most beneficial type of antihypertensive medication in patients with type 1 DM is angiotensin-converting enzyme (ACE) inhibitors (eg, enalapril).



Type 1 Diabetes Mellitus – Review Question

Which of the following clinical findings is most likely to be present in a patient presenting with diabetic ketoacidosis?

- Blood pH of 7.55
- Coma
- Deep and rapid breaths
- Normal to elevated insulin levels
- Seizures



Diabetes & Other Disorders of Endocrine Pancreas 3 of 8: Type 2 Diabetes Mellitus

What Is Type 2 Diabetes Mellitus?

Diabetes mellitus (DM) is a group of metabolic disorders characterized by high blood glucose. Here, we'll address type 2 DM, a disorder that is one of the leading causes of morbidity and mortality in the United States. It leads to severe acute and chronic complications like heart, kidney, and eye disease. It is initially caused by peripheral resistance to insulin, and later by falling insulin levels. The main consequence is high blood sugar levels (hyperglycemia), leading to many of the adverse effects of the disease.

Epidemiology

Type 2 DM is a major public health burden, with 7% to 8.5% of US adults diagnosed. Another 33% have prediabetes, in which glucose levels are high but do not yet meet the standard of DM. Most (90%) cases of DM in the United States are type 2, and it represents nearly all patients diagnosed after age 19 years. In the United States, the risk is higher in Hispanic people, non-Hispanic Black people, Native American individuals (including Alaska Natives), and Native Hawaiian or other Pacific Islander individuals.

Knowledge Check

Question: What percentage of US adults have type 2 DM?

Answer: The percentage of US adults with type 2 DM is 7% to 8.5%.

How Do Patients With Type 2 Diabetes Mellitus Present?

A typical patient with type 2 DM is older than age 40 years and obese with predominately abdominal (visceral) fat. There are many exceptions to this, however, and children now develop type 2 DM at alarmingly high frequencies. Unlike patients with type 1 DM, who are usually symptomatic at presentation, patients with type 2 are most often diagnosed by **lab criteria**, before symptoms develop. Type 2 is caught more quickly because the disease develops more slowly than type 1, and blood glucose is now routinely measured in medical practice.



Knowledge Check

Question: What is the typical age of presentation of patients with type 2 DM?

Answer: The patient is usually older than 40 years of age when presenting with symptoms or lab findings consistent with type 2 DM.

When patients do have symptoms, those are caused by hyperglycemia. The patient may complain about increased urinary frequency (polyuria), increased thirst (polydipsia) and increased hunger (polyphagia). Polyuria occurs because a serum glucose value higher than 240 mg/dL overwhelms reabsorptive capacity of the renal proximal tubules, leading to glucose spilling into the urine. This draws water into the tubule and increases urine volume. Polydipsia is the patient's attempt to keep up with the water loss and thirst by drinking lots of fluids. Patients with type 2 DM have often been gaining weight, unlike the weight loss typically seen in type 1.

Knowledge Check

Question: What are the three P's of DM?

Answer: Polyphagia, Polyuria, and Polydipsia

What Is the Pathophysiology of Type 2 Diabetes Mellitus?

Type 2 DM is characterized by **insulin resistance** in the beginning, then later by high glucagon levels and a deficiency of insulin secretion. The contribution of each factor varies in each patient. These abnormalities are caused by a mix of genetic and environmental factors, including obesity.

Risk Factors

Obesity is a major risk factor, especially in patients with excess intra-abdominal fat (commonly referred to as visceral fat). Other modifiable risk factors include sedentary lifestyle, diet, smoking, alcohol consumption, and sleep deprivation. Some of these overlap with psychosocial, socioeconomic, and behavioral risk factors.

A patient's risk of developing type 2 DM is greatly increased by having a parent who has been diagnosed with type 2 DM—inheritance plays a greater role in type 2 DM than in type 1 DM.

Disease Mechanism

The earliest underlying cause for the development of type 2 DM is **insulin resistance**, followed by inadequate pancreatic insulin secretion and elevated glucagon levels. Other factors that may contribute to type 2 DM include excessive hepatic glucose production, abnormal fat metabolism, systemic low-grade inflammation, and genetics.

Insulin Resistance. Insulin is a hormone produced by the pancreatic β cells and secreted in response to elevated blood glucose after we eat. Insulin-receptor signals mediate the insertion of glucose transporter type 4 onto cell surfaces, allowing glucose to enter into the cells (mainly liver and muscle cells) and thereby lowering blood glucose levels.

In insulin resistance, insulin's ability to shift sugar from the blood into the cells is limited because of multiple insulin-related changes, which largely occur in the liver, adipocytes, and muscles. These changes lead to hyperglycemia even as the pancreas pumps out more and more insulin to try to regulate the body's increased blood glucose levels.

Why do our cells become insensitive to insulin? The causes have been shown to be linked to genetics, obesity, and sedentary lifestyle:

- Elevated free fatty acids in the blood (which blunt the cell's insulin response)
- Reduced number of insulin receptors
- Cytokines secreted by adipocytes such as tumor necrosis factor α
- Deficiency of leptin, the hormone that signals us to stop eating when we are full
- High cortisol levels seen in stress and abnormalities in cyclic secretion of cortisol by the adrenal gland

Metabolic syndrome (also known as insulin resistance syndrome) is an important component of the insulin resistance in many patients with type 2 DM. It is defined by the presence of any three of the following five traits:

- Abdominal obesity (also known as visceral fat)
- High levels of serum triglycerides
- Low levels of serum HDL cholesterol
- Hypertension
- Hyperglycemia

Metabolic syndrome is strongly associated with the development of type 2 DM as well as with cardiovascular disease. Although obesity is a strong risk factor for type 2 DM and metabolic syndrome, it is important to note that the criteria for metabolic syndrome are not focused on body mass index (BMI) or total body weight, but rather abdominal obesity.



Knowledge Check

Question: What kind of fat distribution is relevant to the development of metabolic syndrome?

Answer: Abdominal or visceral fat is associated with metabolic syndrome.

<https://rossmed.instructure.com/courses/3672/pages/fusion-session-|-workshop-diabetes-and-other-disorders-of-endocrine-pancreas-part-3-of-8-type-2-diabetes-mellitus-2>

Inadequate Pancreatic Insulin Secretion. Although insulin resistance is fundamental to type 2 DM, most patients would not develop overt disease if they could produce enough insulin to compensate. Patients with longstanding type 2 DM cannot respond in this way, and if their β cells cannot increase their insulin levels enough, hyperglycemia develops (Figure 1).

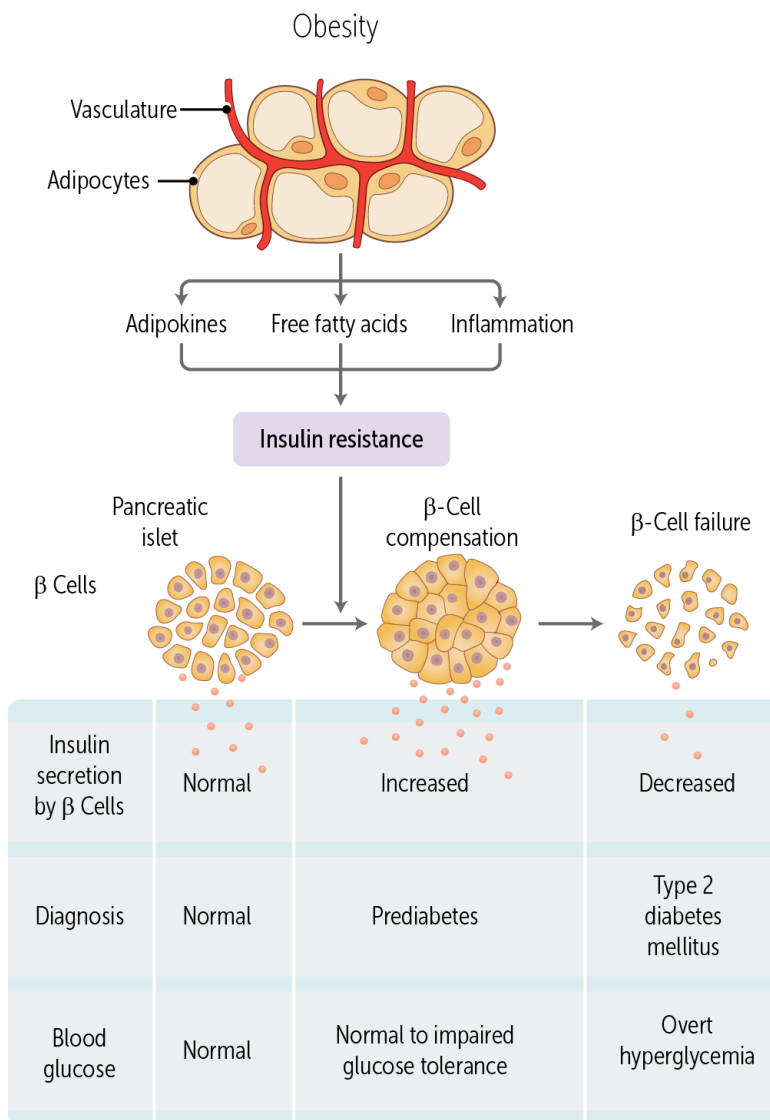


Figure 1. Image Credit: ©ScholarRx, used with permission.

This is important in therapy, because some medications increase insulin sensitivity, whereas others increase pancreatic insulin release.

As shown in Figure 1, insulin resistance initially leads to β-cell hypertrophy with mildly elevated blood glucose levels and increased insulin secretion. This condition is termed **prediabetes**, because patients do not yet meet diagnostic criteria for type 2 DM (see diagnostic criteria below). Eventually their serum glucose rises into diabetic range, and they are diagnosed with type 2 DM. Later, β-cell failure occurs, and these patients with type 2 DM will require insulin therapy.

Elevated Glucagon Levels. Patients with type 2 DM demonstrate high levels of the hormone glucagon. Glucagon is the anti-insulin hormone that releases glucose from its storage molecule, glycogen. In type 2 DM, glucagon's relationship with insulin is



disordered, leading to abnormally high levels of glucagon and increasing the serum glucose.

Knowledge Check

Question: What is the main underlying cause of type 2 DM?

Answer: *Insulin resistance is the main underlying cause of type 2 DM.*

Acute Complications

Patients with type 2 DM may develop **hyperosmolar hyperglycemic state**. This is a marked elevation of serum glucose, often to more than 600 mg/dL, accompanied by massive osmotic diuresis, volume depletion, electrolyte disorders, and hypotension. It is often precipitated by an infection, myocardial infarction, or other acute illness. It may be the initial presentation of type 2 DM, especially in patients who do not receive regular medical care.

Patients with DM are at higher risk for various **infections**, including fungal infections like thrush (oral/tongue) and candida vaginitis.

Chronic Complications

Patients with either type 1 or type 2 DM may develop one or more chronic complications. These complications are usually more prevalent and severe if glucose control has been poor, and they are often caused by the **nonenzymatic glycation of proteins** that occurs when serum glucose levels are chronically high. Glycated proteins are proinflammatory and thicken the walls of the small and large blood vessels, leading to microangiopathy, macroangiopathy, and ischemia. Common chronic complications include:

- Retinopathy, which can lead to blindness
- Diabetic kidney disease, which can lead to nephropathy and the need for dialysis or transplant
- Autonomic neuropathy, causing orthostatic hypotension, gastrointestinal (GI) dysfunction (nausea, vomiting, diarrhea), and urinary retention or incontinence
- Accelerated atherosclerosis, with higher risks of coronary artery disease, stroke, and peripheral artery disease
- Peripheral neuropathy with loss of sensation, which, together with ischemia of the feet, can lead to diabetic foot ulcers (Figure 2)



Figure 2. Image Credit: Courtesy of Wikimedia Commons user Intermedichbo.

Knowledge Check

A patient with long-standing untreated type 2 DM and blurred vision may have which condition?

Answer: A patient with long-standing untreated type 2 DM and blurred vision may have diabetic retinopathy.

How Do We Diagnose Diabetes Mellitus Type 2?



A diagnosis of DM (both type 1 and type 2) is typically made with one or more blood glucose tests (Table 1).

Table 1

Test	Cutoff value	Description
Fasting blood glucose	≥ 126 mg/dL	Requires two positive test results
Glycated hemoglobin (HbA _{1c})	$\geq 6.5\%$	Used more frequently with patients with type 2 DM; value $< 6.5\%$ does not exclude the diagnosis of type 1 DM
Random venous blood glucose	≥ 200 mg/dL	Used only for patients exhibiting symptoms of hyperglycemia
Oral glucose tolerance test	Serum glucose ≥ 200 mg/dL 2 h after the patient ingests glucose	Not frequently used now

The hemoglobin A_{1c} test measures the percentage of red blood cells (RBCs) that are glycated. HbA_{1c} levels increase as glucose levels increase and show how well controlled the patient's blood sugar levels have been over the past 3 months.

If a patient has an abnormal value in any of the four tests shown in the table, the diagnosis of DM should be high on the differential and correlated clinically. In the absence of unequivocal hyperglycemia, results should be confirmed by repeat testing.

Knowledge Check

Question: What are the four tests used to diagnose DM?

Answer: Fasting blood glucose, HbA1c, random venous blood glucose, and oral glucose tolerance tests are used to diagnose DM.

Distinguishing Type 1 from Type 2 Diabetes Mellitus

The clinical distinction between the two types of DM is usually straightforward, given the differences in age, tendency to ketoacidosis, and body weight. But in some children, in whom type 1 DM is more common, the diagnosis may be uncertain after an initial evaluation. Two lab tests are useful in these circumstances.

C-Peptide Levels. C-peptide is a protein that is cleaved from proinsulin as it becomes insulin (Figure 3).

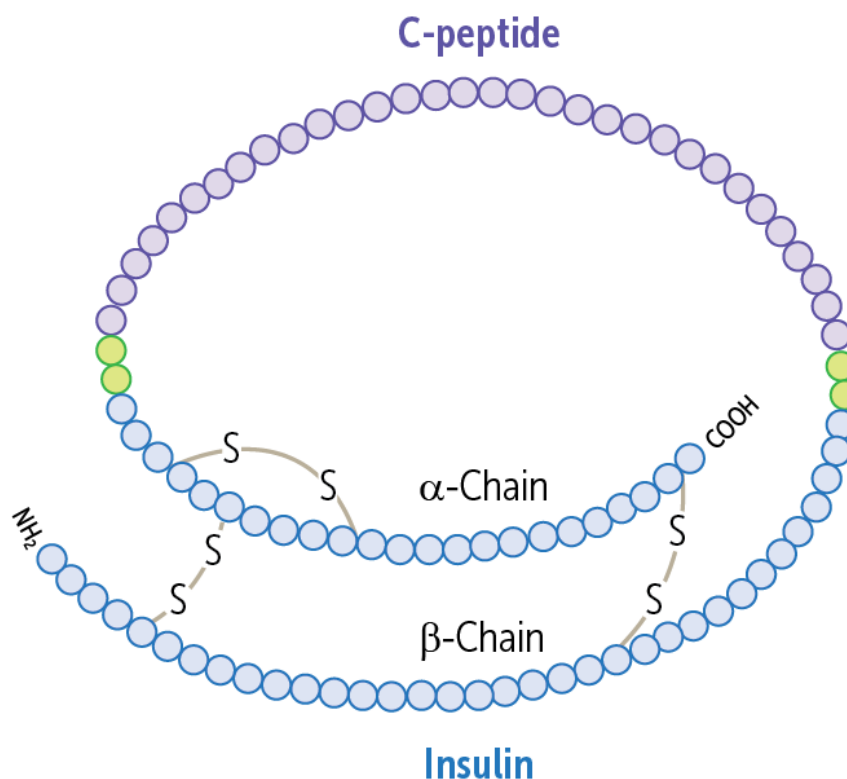


Figure 3. Image Credit: ScholarRx, used with permission.

C-peptide is stored with insulin in the pancreatic β cells, and it is released when insulin is released. A patient with type 2 DM still has endogenous insulin (ie, produced by the cell's protein-manufacturing machinery) and will have C-peptide in their bloodstream, whereas



a patient with type 1 DM will have little or no C-peptide because they make very little insulin.

Knowledge Check

*Question: C-peptide helps you **SEE** how much insulin is in the body.*

Answer: C-peptide diagnoses type 2 DM.

Knowledge Check

Question: Why is C-peptide low or absent in patients with DM type 1?

Answer: C-peptide is cleaved from proinsulin as it becomes insulin; because patients with type 1 DM do not produce insulin, they do not produce C-peptide.

Autoantibodies. Type 1 DM often presents with autoantibodies directed against various cells of the endocrine pancreas. Some examples include glutamic acid decarboxylase antibodies and islet cell cytoplasmic autoantibodies. These antibodies are present in more than 85% of individuals with type 1 DM, but only 5% to 10% of patients with type 2.

How Can We Prevent and Manage Type 2 Diabetes Mellitus?

Unlike type 1 DM, type 2 can be prevented. Once the disease is established, treatment is very different in the two conditions, but the goal is the same: decrease serum glucose, avoid hypoglycemia, and prevent acute and chronic complications.

Prevention

Patients with **prediabetes** have a high risk for developing type 2 DM. It's important to identify these patients early to prevent progression to type 2 DM. Cardiovascular disease can develop even before overt hyperglycemia occurs. Although they do not yet meet diagnostic criteria for type 2 DM, they will have one or more of the following lab results:

- Hemoglobin A_{1c}: 5.7%-6.4%
- Impaired fasting glucose: fasting plasma glucose of 100-125 mg/dL
- Impaired glucose tolerance: 2-hour plasma glucose value of 140-199 mg/dL during a 75-g oral glucose tolerance test

Screening with one of these tests is typically done in adults ages 40 to 70 years. In younger patients, screening should be considered if an individual has a BMI higher than 25 kg/m² and at least one of the following risk factors: hypertension, hyperlipidemia,

positive family history, polycystic ovary syndrome, or gestational diabetes (diabetes in pregnancy).

When identified, patients considered prediabetic should be encouraged to control serum lipids and blood pressure (BP), to remove other cardiac risk factors (eg, by stopping smoking), and to lose weight.

Knowledge Check

Question: What is the fasting blood glucose value for a healthy individual without pre-DM or DM?

Answer: The fasting blood glucose value for a healthy individual is less than 100 mg/dL (5.6 mmol/L).

Management: Lowering the Glucose

As in type 1 DM, management is a mix of lifestyle changes, medications, and continuous monitoring. It can be divided into glucose control and reduction of vascular complications.

Education and Lifestyle Modifications. Intensive patient education is important in DM management, given all the changes in lifestyle and medications that are needed. **Weight loss** for obese patients can lower blood glucose and reduce the need for medication. Efforts to reduce body weight should include exercise and diet changes. Diets for patients with type 2 DM consist of high amounts of complex carbohydrates and fiber and low amounts fat, sucrose, fructose, and other refined carbohydrates.

Antihyperglycemic Drugs. Medications will be required for most patients with type 2 DM to control their serum glucose. These drugs lower glucose levels through a variety of mechanisms, including decreasing insulin resistance, increasing insulin release, increasing urinary glucose excretion, and decreasing carbohydrate absorption in the GI system. Medications can be oral or injectable. The first one prescribed is usually **metformin**, which is associated with weight loss, lipid reduction, and a low incidence of hypoglycemia. Evidence suggests that metformin and gliptins may aid in weight loss for patient with type 2 DM and obesity.

Insulin may be required to normalize the blood glucose in type 2 DM but should be avoided until one to three oral agents have failed to normalize hyperglycemia or there is very severe, symptomatic hyperglycemia. This avoidance is because in type 2 DM, insulin promotes weight gain and more insulin resistance. Insulin also increases the risk of hypoglycemic episodes.

Monitoring Glucose Control. The glycated hemoglobin (**hemoglobin A_{1c}**) level is the most important single test for monitoring glucose control. The target goal should be below **7.0%** in most patients. It should be checked every 3 months.



Home monitoring of blood glucose levels is necessary for patients taking insulin, but most other patients will not usually need to do this. Home monitoring is advisable when oral antihyperglycemic drugs are changed or if the patient is prone to hypoglycemic episodes.

Monitoring and Preventing Diabetic Complications

All patients should be routinely monitored for chronic complications of DM. This includes:

- **Hypertension:** BP should be less than 140/90 mm Hg and less than 130/80 mm Hg if the patient's cardiovascular risk is high. An angiotensin-converting enzyme (ACE) inhibitor is the best drug for initial hypertension treatment.
- **Hyperlipidemia:** Statin therapy should be used for all patients with DM and cardiovascular disease or high cardiac risk factors. The American Diabetes Association also recommends statin therapy for diabetic patients ages 40 to 75 years even without these risk factors or hyperlipidemia.
- **Retinopathy:** Annual ophthalmology exams are recommended.
- **Nephropathy:** Annual determination of urine albumin, an early sign of diabetic kidney disease, should be taken. An ACE inhibitor should be started if urine albumin exceeds 30 mg/24 hours or 30 mg albumin per gram creatinine on a spot urine specimen.
- **Neuropathy:** Annual in-office exams of the feet should be done for ulcers and for sensory loss; sensory loss should be determined with pinprick, vibratory, and microfilament testing.

Drugs for Reducing Complications. Because both hypertension and hyperlipidemia are strong risk factors for diabetic complications, they should be aggressively managed.

- **BP control** should target a BP under 140/90 mm Hg, or 130/80 mm Hg if there is concurrent cardiovascular disease. Multiple drugs may be needed. An ACE inhibitor like enalapril is often used for this purpose, given its beneficial effects on the heart and kidney. ACE inhibitors should always be used for diabetic kidney disease. This is because these drugs delay the progression, in part because they reduce glomerular capillary pressure by dilating the efferent arteriole.
- **Statin therapy** is used to treat hyperlipidemia and is needed in all adult patients with DM who have high cardiovascular risk factors or who are at least 40 years of age.



Case Connection

Thinking back to TC, how do you help her manage her DM in a cost-effective way?

You tell TC that with proper attention to weight management and diet, she may be able to manage her DM without expensive medications. You tell her that the first-line medication, metformin, is generic and low cost, as are some statins and ACE inhibitors. She needs to plan for a yearly retinal exam and for regular tests to monitor her kidneys. "What about my own testing supplies?" TC asks. You tell her that if her hemoglobin A1c remains in a reasonable range with the measures you've discussed today, she might not need to routinely check her blood sugar at home, because home monitoring doesn't seem to improve outcomes.

Summary

- Type 2 diabetes mellitus (DM) is a common metabolic disease characterized by a combination of insulin resistance and inadequate insulin secretion, leading to hyperglycemia.
- Type 2 DM typically affects adult patients, but children can also get it.
- Patients with type 2 DM are usually older than age 40 years and obese with predominantly abdominal fat.
- Most patients with type 2 DM are diagnosed by lab criteria prior to development of symptoms.
- Insulin resistance is related to obesity, genetics, and abnormal cytokine production.
- Patients with type 2 DM may also have decreased insulin secretion and elevated glucagon levels.
- A strong association between metabolic syndrome and type 2 DM has been demonstrated, as has a strong association between metabolic syndrome and cardiovascular disease.
- Both type 1 and type 2 DM are diagnosed by the same lab criteria.
- C-peptide levels can be useful to distinguish type 1 (low or no C-peptide) from type 2 DM.
- Patients with type 2 DM lack the classical antibodies associated with type 1 DM, such as glutamic acid decarboxylase antibodies and islet cell cytoplasmic autoantibodies.
- Prediabetes is a state in which a patient does not meet the cutoff to be considered diabetic but is at risk for doing so; such patients should be monitored closely.
- Complications include retinopathy, nephropathy, neuropathy, cardiovascular disease, and infections.



- Patients should be followed up with hemoglobin A1c testing.
- Treatments include weight loss, diet, antihyperglycemic drugs (eg, metformin), and insulin (if required).
- Blood pressure and serum lipid levels should be well controlled; angiotensin-converting enzyme inhibitors are the preferred antihypertensive agent in most cases.

Type 2 Diabetes Mellitus – Review Question

Which test results would best confirm a diagnosis of type 2 diabetes mellitus in a symptomatic patient?

- Blood hemoglobin A_{1c} result of 4.1%
- Overnight monitoring of glucose levels at an accredited center
- Random blood glucose test showing a value of >200 mg/dL
- Serial glucose blood tests for a week
- Two fasting blood glucose test results >110 mg/dL

Diabetes & Other Disorders of the Endocrine Pancreas Part 4 of 8: Diabetic Ketoacidosis

Introduction

Case Connection

BW, a 21-year-old college student, was brought to the emergency department by his roommate during finals week. The roommate had found BW puking in the bathroom. He reports that BW was confused and had wet himself. He had not consumed any alcohol because of an exam scheduled the next day. You enter the room to find BW, groaning and clutching his stomach. He is breathing deeply, with a respiratory rate of 24 breaths/min, and you note a fruity breath odor. His blood glucose is greatly elevated; he has low sodium, high potassium, and a high level of ketones in his blood and urine. His admission orders include intravenous fluids and an insulin drip. Also included is an order for supplemental potassium chloride if BW's potassium level goes below 5.3 mg/dL, even though BW's current level is much higher than normal.

What diagnosis would make you consider how glucose and insulin will affect BW's potassium level? Consider your answer as you read, and we'll revisit BW at the end of the brick.

What Is Diabetic Ketoacidosis?

Diabetic ketoacidosis (DKA) is one of the most dreaded acute complications of diabetes mellitus (DM). This metabolic syndrome leads to severe hyperglycemia and metabolic acidosis (low blood pH) and is a medical emergency. DKA occurs because of insufficient insulin, making it much more common in patients with type 1 DM, who have no endogenous insulin, than in those with type 2 DM, who typically have some functioning islet cells. DKA is often the initial presentation of type 1 DM: the patient, previously undiagnosed with DM, comes to the emergency department severely ill.

DKA is often the first sign that a patient has undiagnosed type 1 DM. Patients with known disease who have missed doses of insulin may also develop DKA, especially if they have physical stressors like a myocardial infarction (MI) or infection. Other risks for DKA include pregnancy (stress, increased insulin demand) and the use of some medications, including glucocorticoids.

Up to 2% of patients with DKA may die from the disorder, so it's imperative that clinicians understand the complex multistep treatment.



How Do Patients With Diabetic Ketoacidosis Present?

When patients with DKA arrive at the emergency department, they are clearly ill, but the symptoms may be nonspecific (eg, vomiting, diarrhea, disordered breathing, and mental status changes—even unconsciousness). There may be a recent history of acute illness like pneumonia, urinary infection, or MI. These stressors increase insulin demands and therefore the risk of DKA. The ketones produced in DKA may cause a fruity breath, and the patient feels weak with flu-like symptoms.

But this is only the start. DKA affects many other body systems.

Urinary Symptoms

Glucose is filtered into the renal tubules, pulling water out of the cells and causing high urine volume (**polyuria**); this set of events is called osmotic diuresis. As we'll see, electrolytes like sodium and potassium follow, leading to severe **volume depletion**.

Cardiovascular Symptoms

Volume depletion from osmotic diuresis causes low cardiac preload, which leads to decreased cardiac output; decreased cardiac output results in **dizziness** and **tachycardia**. Dry mucous membranes and decreased skin turgor (skin tenting) are also seen.

Gastrointestinal Symptoms

Abdominal pain, **nausea**, and **vomiting** are classic symptoms of DKA. These symptoms may be caused by the ketoacidosis and often resolve with its correction. Vomiting worsens the dehydration already present. Elevated prostaglandins may also play a role in abdominal pain; they are released from overactive adipose tissue and can cause vasodilation of various organs, including those in the gastrointestinal system.

Respiratory Symptoms

In DKA, the buildup of acidic ketones has put the body in a state of metabolic acidosis. This drops the serum pH, often to dangerously acidic levels. Respiratory compensation takes the form of **labored deep breathing** known as Kussmaul respirations, in which the body attempts to get rid of the excess CO₂ from acidosis by breathing more deeply and rapidly.

Knowledge Check

Question: Why do patients in DKA breathe deeply?

Answer: The deep breaths with DKA, called Kussmaul respirations, are a form of hyperventilation to compensate for metabolic acidosis.

What Is the Pathophysiology of Diabetic Ketoacidosis?

DKA originates with the lack of adequate insulin and increased concentrations of insulin counterregulatory hormones; the condition is then compounded by volume depletion. DKA is common when the body has increased insulin demands. Common triggers include acute illnesses (such as ischemic heart disease, stroke, sepsis, pancreatitis, etc), medications (such as glucocorticoids), dehydration, cocaine use, and poor adherence to the insulin regimen.

Knowledge Check

Question: What are the risk factors for DKA?

Answer: The 5 Is: Infection, Infarction, Insulinopenia, Iatrogenic causes (eg, medications), and Infant related (eg, pregnancy)

Low insulin levels lead to hyperglycemia, but stress causes the secretion of insulin counterregulatory hormones (hormones that normally oppose the actions of insulin). These stress-induced hormones include **glucagon**, **cortisol**, **catecholamines**, and **growth hormone**. They activate hepatic gluconeogenesis, increasing the serum glucose even more and causing protein breakdown (proteolysis) in muscle cells.

Glucagon increases the availability of free fatty acids. It also stimulates the breakdown of glycogen in the liver (increasing the serum glucose). It furthermore causes glycogen breakdown in the muscle, leading to the release of both lactic acid, which contributes to acidosis, and prostaglandins, which may play a role in abdominal pain and other symptoms.

Knowledge Check

Question: How can you predict glucagon's effects?

Answer: Glucagon's effects are what you would expect if the GLUCose is GONe.

Ultimately, the combination of low-insulin and high-glucagon counterregulatory hormones leads to the production of the ketone bodies **acetoacetate** and **β -hydroxybutyrate**, which are the root causes of DKA (Figure 1).

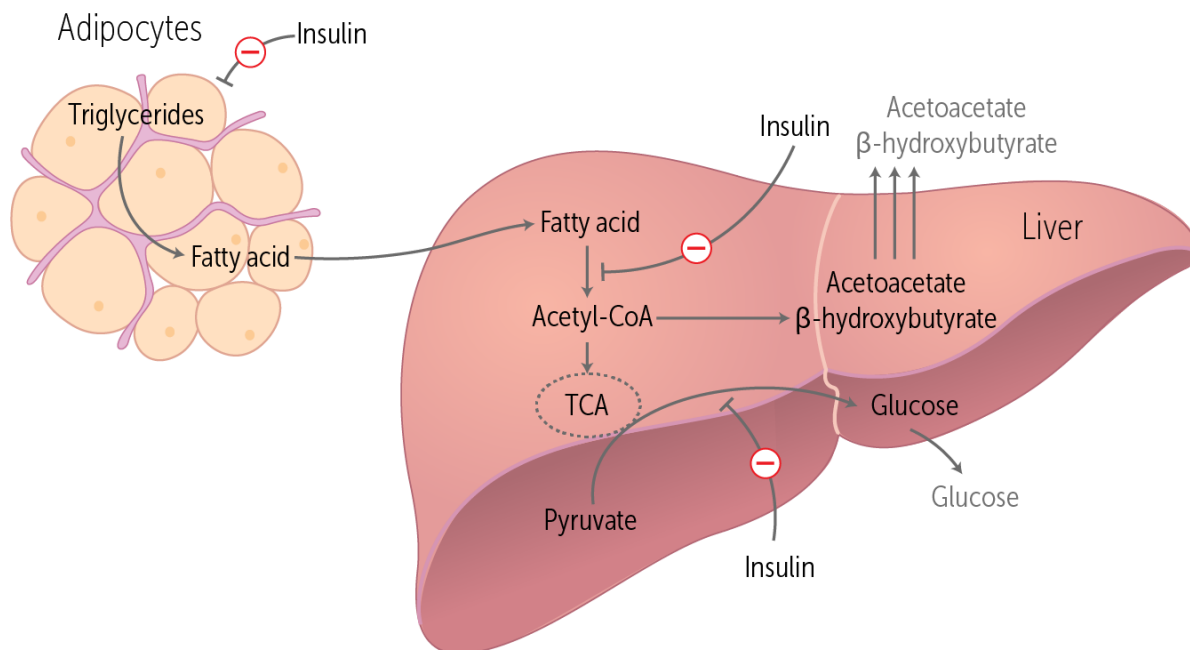


Figure 1. Image Credit: ScholarRx, used with permission.

Ketoacids are normally formed from free fatty acids in the liver when the body is in a state of starvation and needs emergency energy. In DKA, the low insulin and high glucagon are signals to the liver that it should make ketones, even though the patient is not actually starving. As seen in Figure 1, the liver converts excess acetyl coenzyme A (normally a substrate for the tricarboxylic acid [TCA] cycle) into ketone bodies as an alternative source of energy. Normally, many different cell types can use ketones for energy in situations when glucose is unavailable. In DKA, the ketones are produced in such large amounts that they cannot be used up, so they circulate at high levels and cause **metabolic acidosis**. Because the metabolic acidosis is caused by the gain of an acid that is not normally present in the serum (ie, ketone bodies), the specific term for it is **high anion gap metabolic acidosis**. The increased lactic acid that we mentioned earlier also contributes to this metabolic acidosis. The blood pH can fall to very low levels in



DKA, sometimes to less than 7.0 (normal is 7.4), which is the point at which many enzymes stop functioning.

Volume Depletion

The high-glucose blood is filtered into the renal proximal tubules, but the amount of glucose is too great for the kidney to reabsorb. As a result, the glucose concentration (and osmolarity) in the tubule is high, causing water to be pulled in from the cells. This process, called **osmotic diuresis**, increases urine volume, leading to excess urination (polyuria). Polyuria eventually leads to dehydration and polydipsia (excessive thirst).

Electrolyte Abnormalities

The osmotic diuresis of DKA results in a variety of electrolyte abnormalities. Sodium, phosphate, and potassium are lost in the urine, depleting the patient of these electrolytes. The low total sodium in the body leads to signs of volume depletion like hypotension and tachycardia. Phosphate depletion can lead to muscle weakness and low levels of 2,3-bisphosphoglyceric acid (2,3 BPG) in RBCs, which limits their ability to release oxygen to tissues. This contributes to tissue hypoxia.

What about potassium? There the story is a bit more involved. With all this potassium lost in the urine, you would think the patient with DKA would be **severely depleted in potassium**, and you would be right. But you would also guess that would make the patient hypokalemic, but this is most often not the case. When initially presenting with DKA, a patient may have **normal to high serum potassium levels (hyperkalemia)**, despite the low total body potassium. This is because most potassium is inside the cells. During DKA, much of it shifts out into the blood, causing hyperkalemia. This happens for three main reasons:

- Insulin normally stimulates the Na^+/K^+ -ATPase to pump potassium from the blood into cells, so a lack of insulin means this pump is inactive, raising the potassium level in the blood.
- Hyperglycemia increases the serum osmolarity, pulling water out of all cells, and the potassium follows along (osmotic shift).
- To buffer the decreased serum pH, the membrane H^+/K^+ exchanger brings extracellular H^+ into the cell and pumps K^+ out.

So, the total level of potassium is low in the patient's body, but the initial level in the serum is high. Remember that—we'll come back to it when we discuss treatment.

Figure 2 summarizes all the physiologic changes.

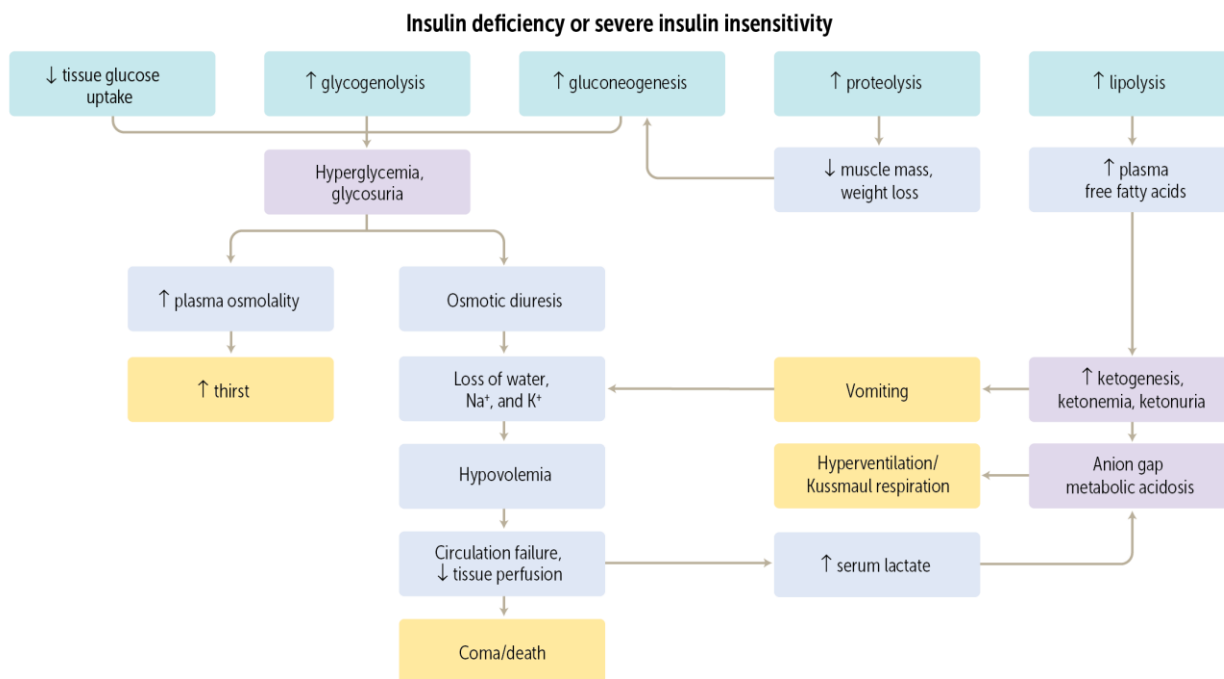


Figure 2. Image Credit: ScholarRx, used with permission.

Knowledge Check

Question: What is the serum potassium level in most patients presenting with DKA?

Answer: In most patients presenting with DKA, the serum potassium level is normal to elevated (hyperkalemia).

How Do We Diagnose Diabetic Ketoacidosis?

Although DKA is often recognizable from presenting symptoms (hypotension, abdominal pain, fruity breath, and deep respirations), the diagnosis is best made with laboratory findings.

Because medical complications like infections and MIs may precipitate DKA, all patients with DKA need a thorough history and physical exam as well as a complete blood count and metabolic panel. Urinalysis is done to look for infections as well as to look for the ketone evidence of DKA. An x-ray of the chest is routine to look for pneumonia.



Laboratory Values

DKA presents with a host of lab abnormalities, but each one has a logical explanation.

Metabolic Acidosis. First and foremost, DKA is an acid-base disturbance, so arterial blood gas readings will show metabolic acidosis: blood pH less than 7.40 (often <7.10) with a low serum bicarbonate concentration (high acid lowers bicarbonate buffer) and low partial pressure of carbon dioxide (respiratory compensation for acidosis by Kussmaul respirations). In high anion gap metabolic acidosis, high levels of unmeasured anions (ketones as well as some lactic acid) are present in the serum. The ketones, β -**hydroxybutyrate** and **acetoacetate**, can be measured in the urine or in the serum.

Knowledge Check

Question: What is the primary acid-base disturbance in DKA?

Answer: The primary acid-base disturbance in DKA is a high anion gap metabolic acidosis.

Hyperglycemia. DKA is also characterized by **hyperglycemia**. The degree of hyperglycemia ranges from mild to severe (the glucose levels can be anywhere from 250-600 mg/dL). This occurs because the patient does not have enough insulin to transport glucose into cells.

Hyponatremia. The high serum glucose causes **high serum osmolarity**, which draws water out of cells and thus dilutes the serum and leads to hyponatremia. Serum sodium is usually more than 125 mEq/L. However, unlike in syndrome of inappropriate antidiuretic hormone (SIADH), in which serum osmolarity is low, in DKA, serum osmolarity is high; hence, the hyponatremic symptoms seen in SIADH are not seen in DKA.

Potassium. As mentioned, patients often present with **normal serum potassium or hyperkalemia**, because even though total body stores of potassium are low, potassium has shifted out of cells and into the blood. The level will fall during treatment with insulin (which will transport the potassium back into the cells), so patients will often develop **hypokalemia** with continued insulin treatment. Here, potassium repletion is needed because overall stores are decreased.

Other Electrolytes. Hypomagnesemia and hypophosphatemia may also be present in DKA because of osmotic diuresis and may require repletion.

Acute Kidney Injury. DKA is usually accompanied by prerenal acute kidney injury resulting from dehydration and low renal blood perfusion. The serum urea nitrogen (BUN) and serum creatinine will be elevated, usually with a BUN:creatinine ratio of more than 20. This problem corrects with volume repletion.



How Do We Manage Diabetic Ketoacidosis?

DKA must be managed immediately because the electrolyte abnormalities can be fatal. Common causes of death are cerebral edema from rapid fluid shifts and cardiac arrest from electrolyte abnormalities. DKA therapy is a big challenge for hospital clinicians, and the patient often needs to be monitored in the intensive care unit. Prompt treatment normally prevents permanent damage from the disorder.

Volume Repletion

The first goal for therapy in DKA is volume repletion. Dehydration and volume loss in DKA can be so severe that patients require 5 to 10 L of fluid resuscitation. Patients should be treated with **normal (isotonic) saline** because it largely stays in the extracellular compartment. Giving fluids increases blood pressure and the glomerular filtration rate, improving kidney function. This helps the kidney excrete the ketones.

Knowledge Check

Question: What is used for volume repletion in the initial treatment of DKA?

Answer: In the initial treatment of DKA, volume repletion is achieved with normal (isotonic) saline.

Insulin

In addition to fluid resuscitation, a patient with DKA also needs intravenous (IV) insulin. Insulin will shut off ketone production, allowing the ketones to be excreted in the urine, correcting the metabolic acidosis. Insulin will inhibit glucagon production, reduce glucose formation, and store glucose as glycogen, all of which will lower the serum glucose level.

The insulin must be administered intravenously to provide better control than the slower method of subcutaneous administration. IV insulin should be given until the serum ketones are gone (and the acidosis resolves). The insulin will also lower the blood glucose concentration, and that may happen faster than the fall in ketones. Once the serum glucose level drops to less than 200 mg/dL, dextrose should be added to the IV fluids to prevent hypoglycemia.

When the ketones are no longer present in the blood, and if the patient can eat, you can transition to subcutaneous insulin.

Potassium and Phosphate

Potassium is one of the most crucial electrolytes to monitor in the treatment of DKA. As mentioned, patients are often hyperkalemic at presentation of DKA and may have associated arrhythmias, seen as peaked T waves on ECG (Figure 3).

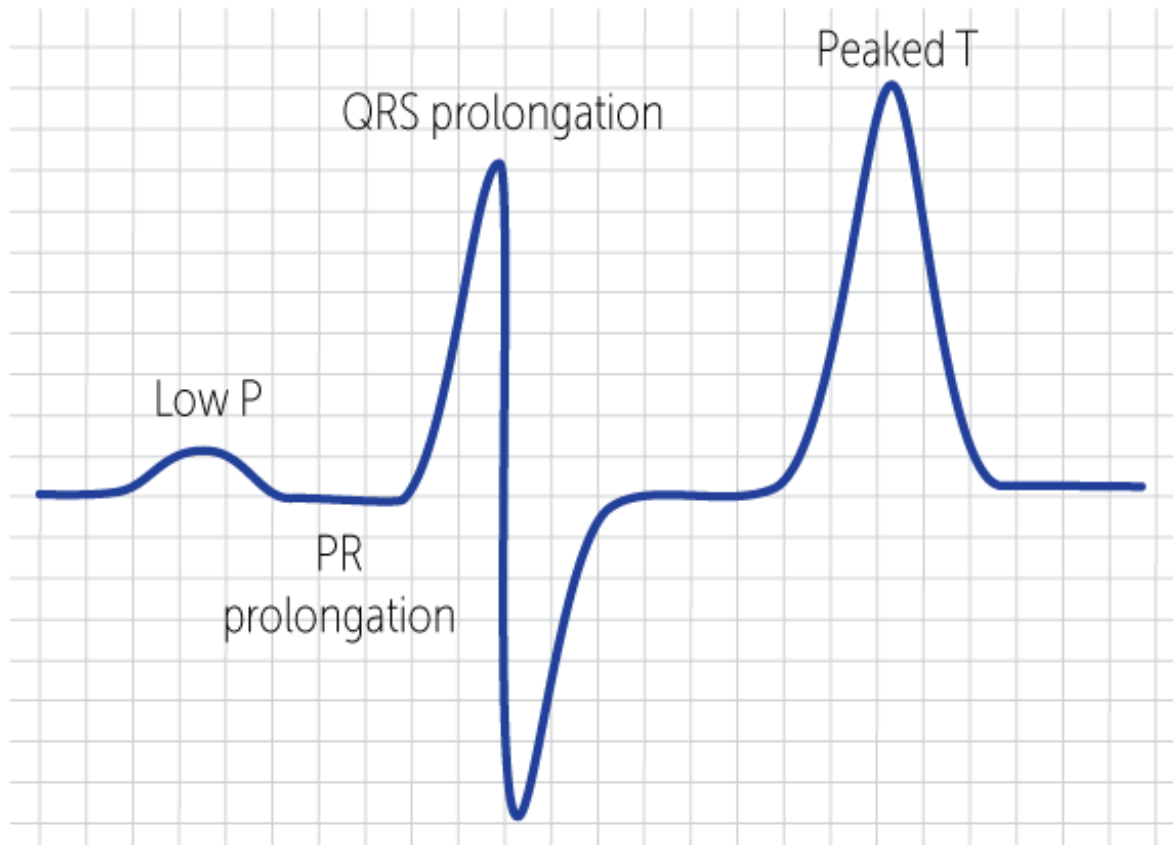


Figure 3. Image Credit: ScholarRx, used with permission.

Hyperkalemia will resolve with insulin treatment. But because the overall potassium stores in the patient's body have been depleted (because of osmotic diuresis), **hypokalemia** can quickly develop after insulin is administered. This can cause life-threatening cardiac arrhythmias, which can be predicted on ECG by U waves (Figure 4).

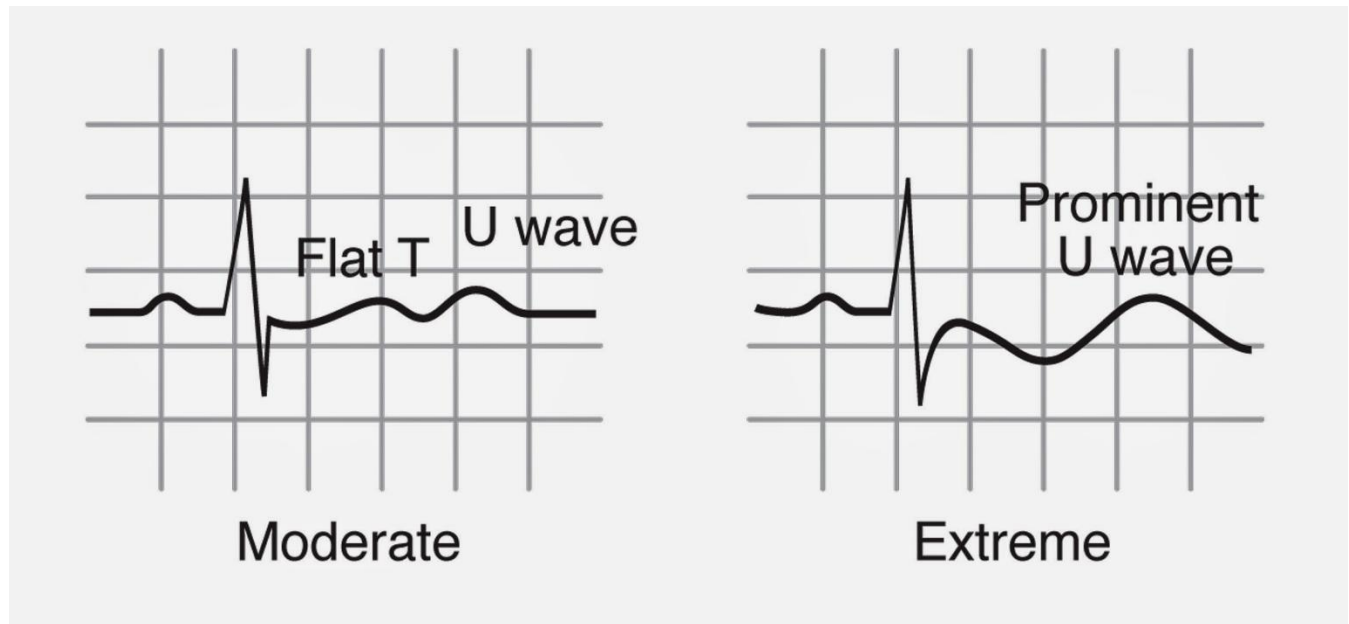


Figure 4. Image Credit: ScholarRx, used with permission.

To avoid hypokalemia during insulin treatment, potassium should be added to the IV fluids when the serum potassium is less than 5.3 mEq/L. That's right; give potassium even when the patient is still hyperkalemic, because you know the serum potassium will keep falling.

Hypophosphatemia and hypomagnesemia also occur with DKA, but phosphate is only administered if the serum phosphate level is less than 1.0 mg/dL. Magnesium can be given if the level is less than 2.0 mg/dL.

Chronic Management

If your patient is having DKA as their initial presentation with type 1 DM, you will train them on diet and insulin administration for when they exit the hospital. Interestingly, some patients will not immediately require insulin after their initial bout of DKA or will require very small doses. This may be because the infection or stress that set off the DKA episode is now resolved, and because they are still making some insulin. This "honeymoon period" may last months or even years. But they should be followed closely, as they will soon need insulin when their endogenous supply is turned off.



Case Connection

Thinking back to BW, what diagnosis makes you consider how glucose and insulin will affect BW's potassium level?

BW exhibits classic signs of DKA, and you remember what happens during treatment. Although serum potassium levels are elevated initially, total body stores are often depleted during osmotic diuresis. The remaining potassium stays in the extracellular fluid, which makes the serum potassium appear high. Once insulin is available, potassium is taken back into the cells, and the serum level may fall precipitously (and dangerously). To maintain the patient's serum level in a safe range, potassium is added to the IV fluids before the patient's serum potassium falls too low.

Summary

- Diabetic ketoacidosis (DKA) is a severe acute complication of diabetes mellitus (DM) that leads to hyperglycemia, blood ketones, and acidosis.
- DKA is more common in type 1 DM but can present in type 2 DM.
- Common risk factors for DKA include lack of insulin, increased levels of stress (eg, infections, myocardial infarction), and increased insulin requirements.
- Common symptoms of DKA include abdominal pain, nausea, vomiting, decreased skin turgor, dry mucous membranes, and loss of mental acuity.
- Kussmaul breathing is the hyperventilation that occurs as patients attempt to get rid of the excess CO₂ from acidosis.
- DKA results from a lack of adequate insulin coupled with the activation of counterregulatory hormones.
- Persistent hyperglycemia in the setting of low insulin levels leads to production of the ketone bodies acetoacetate and β -hydroxybutyrate.
- Osmotic diuresis leads to volume depletion and is a key component in the pathophysiology of DKA.
- DKA is an acid-base disturbance.
- Common laboratory findings include hyperglycemia, high anion gap metabolic acidosis, hyperkalemia, hypomagnesemia, hypophosphatemia, and increased urine and serum ketones.
- Management of DKA includes fluid resuscitation, insulin, and potassium.
- Careful attention must be given to electrolyte abnormalities that may occur during treatment.



<https://rossmed.instructure.com/courses/3672/pages/fusion-session-|-workshop-diabetes-and-other-disorders-of-endocrine-pancreas-part-4-of-8-diabetic-ketoacidosis>

Diabetic Ketoacidosis - Review Question

Which of the following is most likely to occur after the administration of insulin therapy in a patient with diabetic ketoacidosis?

- Magnesium moves out of cells
- Phosphate moves out of cells
- Potassium moves into cells
- Potassium moves out of cells
- Sodium moves into cells



Diabetes & Other Disorders of the Endocrine Pancreas Part 5 of 8: Hyperosmolar Hyperglycemic State

Introduction

Case Connection

JV is a 77-year-old male with type 2 diabetes mellitus (DM) who is living with his daughter after a hip fracture. She brings him to urgent care because he has been refusing his medications and eating poorly for several days. When she tried to check his blood sugar, she couldn't get a reading on her glucometer. In the clinic, his blood glucose and serum osmolarity are high, but ketones are not elevated. You arrange to transfer JV to the hospital for urgent treatment of hyperosmolar hyperglycemic state. His daughter asks, "Can't you just give him insulin here to bring his sugar down?"

What will you tell JV and his daughter about the best way to treat his condition immediately? Consider your answer as you read, and we'll revisit JV at the end of the brick.

What Is Hyperosmolar Hyperglycemic State?

Type 2 diabetes mellitus (DM) is known for associated long-term complications that develop over many years of disease, such as damage to small blood vessels and nerves. But acute complications may also arise, the most dangerous of which is hyperosmolar hyperglycemic state (HHS). The name gives away what happens: **severe hyperglycemia** (often >600 mg/dL) and consequent marked **high blood osmolarity** that causes massive water shifts.

Like its cousin diabetic ketoacidosis (DKA) in type 1 DM, HHS develops in the setting of acute illness or lack of treatment for type 2 DM and can be fatal. So clinicians must be on high alert for HHS in their patients with type 2 DM.

In most cases, a trigger for HHS can be identified. Common triggers include acute illnesses (such as ischemic heart disease, stroke, sepsis, pancreatitis, etc), medications (such as glucocorticoids), dehydration, cocaine use, and poor adherence to diabetes medications.

Knowledge Check

Question: What are some of the major triggers for HHS?

Answer: Common triggers for HHS include acute illnesses (eg, ischemic heart disease, stroke, sepsis, pancreatitis, etc), medications (eg, glucocorticoids), dehydration, cocaine use, and poor adherence to diabetes medications.

How Do Patients With Hyperosmolar Hyperglycemic State Present?

The earliest symptoms of HHS in patients with type 2 DM are like the initial presentation of type 1 DM: **polydipsia**, **polyuria**, and **weight loss**. These symptoms are all caused by severe hyperglycemia, and they develop over several days. Patients develop signs of **volume depletion** caused by osmotic diuresis: decreased skin turgor (skin tenting), dry oral mucosa, low jugular venous pressure, tachycardia, and hypotension.

Patients may develop **neurologic symptoms**, including blurred vision, headache, altered mental status, and seizures. These neurologic symptoms are related to the extent of hyperglycemia and are a consequence of fluid shifts in the brain, as water leaves it because of the high serum osmolarity.

Because stress may set off HHS, the initial clinical presentation may be with an infection such as pneumonia or a myocardial infarction (MI). In addition, because the hyperosmolar state and intravascular volume depletion can trigger thrombosis, patients may present with a thrombotic stroke or new seizures.

Knowledge Check

Question: Why do patients with HHS have neurologic symptoms?

Answer: Patients with HHS tend to have neurologic symptoms because high serum osmolarity causes water to leave the brain.

What Is the Pathophysiology of Hyperosmolar Hyperglycemic State?

HHS typically occurs in older patients with poorly controlled type 2 DM. HHS is characterized by very high blood glucose concentrations, which lead to high serum osmolarity. Notably, significant ketoacidosis is not a feature of HHS, distinguishing it from DKA.

Knowledge Check

Question: What are the major risk factors for HHS?

Answer: The major risk factors for HHS are uncontrolled type 2 DM and advanced age.

Hyperglycemia and Volume Depletion

The pathogenesis of HHS centers on hyperglycemia and hyperosmolarity (Figure 1).

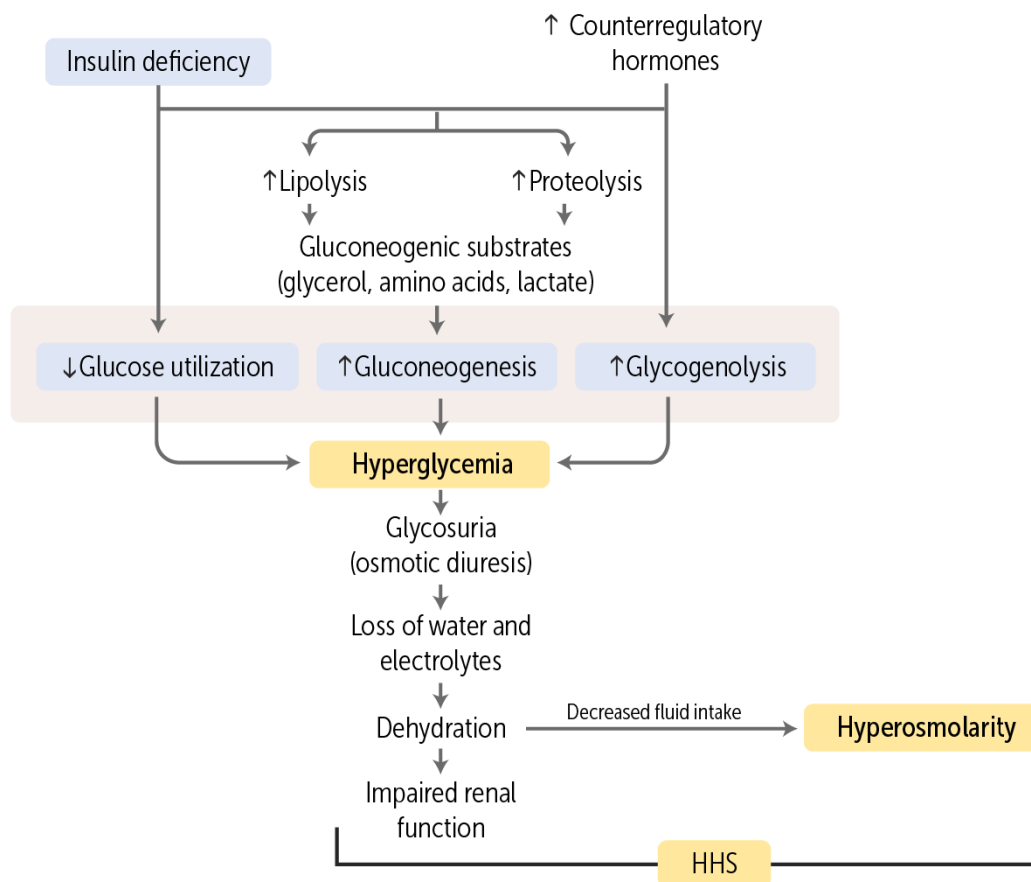


Figure 1. Image Credit: ©ScholarRx, used with permission.

A trigger—such as underlying infection, MI, or dehydration—will cause the release of stress hormones such as catecholamines, cortisol, glucagon, and growth hormone. These counterregulatory hormones act to increase blood sugar levels by accelerating gluconeogenesis and glycogenolysis and decreasing glucose utilization by tissues. This is an adaptive mechanism, as stress often requires extra glucose to be readily available to the muscle (fight-or-flight response). But in HHS, this process causes blood sugars to spike uncontrollably, often exceeding 1000 mg/dL (normal is <100 mg/dL).



The increased glucose levels raise the serum osmolarity, causing water to shift from inside cells into the serum. The glucose is filtered into the urine, where the high osmolarity causes water to be drawn into the tubule, leading to **osmotic diuresis** marked by very high urine output. The proximal renal tubules can typically reabsorb glucose until the blood glucose concentration reaches 270 mg/dL. Over this level, both glucose and water will be excreted. Osmotic diuresis causes volume depletion, leading to dizziness and tachycardia. Sodium and potassium are also lost in the urine during diuresis. The loss of water and sodium causes significant **volume depletion** (often as much as 8 to 10 liters in deficit) as well as **low total body potassium**.

Potassium Abnormalities

With all this potassium lost in the urine, you'd think that the patient with HHS would be severely hypokalemic, but this is not the case. Just like in DKA, when initially presenting with HHS, a patient may often have normal or high serum potassium levels (hyperkalemia), despite the low total body potassium. Why? Recall that most of the body's potassium is inside the cells. In HHS, much of this potassium shifts out of the cells into the blood, causing **hyperkalemia**. These are the two main reasons:

- The severe hyperglycemia increases the serum osmolarity, pulling water out of all cells, and potassium follows along (osmotic drag).
- Insulin normally stimulates the Na^+/K^+ -ATPase to pump potassium into cells from the blood, so a reduced level of insulin means this pump is less active, raising the potassium level in the blood.

So in the HHS patient, the level of total body potassium is low, but the initial level of serum potassium is normal or high. We'll come back to this when we discuss treatment, but first we need to understand why there's no ketoacidosis, unlike in DKA.

Differentiating HHS from DKA

In type 1 DM, DKA is the analogous acute complication. Like HHS, DKA shows hyperglycemia and hyperosmolarity. But it also shows high blood ketones and ketoacidosis—a low blood pH. So why doesn't HHS cause ketones in the blood, like DKA does?

Remember that in type 2 DM, cells are resistant to insulin, but the pancreas may still make insulin. Even a small amount of insulin is enough to suppress lipolysis (free fatty acid production) and the subsequent ketone formation that characterizes DKA, in which insulin is completely absent. However, stress hormones such as epinephrine continue to try to increase the energy available for the body by further raising blood glucose levels. So a patient with HHS will have extremely high glucose and high serum osmolarity, but that patient will usually have no ketones and therefore will lack the severe metabolic acidosis seen in DKA.



Knowledge Check

Question: What are the key metabolic findings in HHS?

Answer: The key metabolic findings in HHS are hyperglycemia and hyperosmolarity with little to no ketosis.

How Do We Diagnose Hyperosmolar Hyperglycemic State?

The diagnosis of HHS is suggested by the initial clinical presentation but is made by the results of blood tests. Typically, a blood glucose level >600 mg/dL, serum osmolarity >320 mOsm/L (normal is 270-290), and the absence of ketoacidosis (no serum ketones, blood pH >7.3) will confirm the diagnosis of HHS. But here's a more detailed list of all the relevant test results:

- **Glucose:** Blood glucose levels are typically extremely high in HHS and can exceed 1000 mg/dL.
- **Osmolarity:** Because of extremely high glucose levels, the serum osmolarity will usually be >320 mOsm/L.
- **Ketones:** Unlike the elevated levels in DKA, blood ketones are usually absent in HHS. There may be trace urine ketones in any dehydrated patient, so the urine test is nonspecific, and a positive urine test does not rule out HHS. The lack of **serum ketones** signifies the absence of metabolic acidosis (elevated anion gap) seen in DKA.
- **Arterial pH:** >7.3 , unlike DKA.
- **Sodium:** The high blood glucose draws water out of cells, diluting the blood and leading to hyponatremia (low serum sodium concentration). The serum sodium in HHS is usually >125 mEq/L, and the hyponatremic symptoms seen in the syndrome of inappropriate antidiuretic hormone (SIADH) are not present, because the serum osmolarity is high.
- **Potassium:** Hypokalemia, normal potassium, or hyperkalemia may be present initially, depending on the patient's insulin levels and degree of dehydration. Total body potassium is uniformly decreased, so potassium supplementation will be needed initially or during or after treatment, whenever the serum level falls.
- **Blood urea nitrogen/creatinine:** These results are elevated because of massive dehydration, leading to prerenal acute kidney injury. Volume repletion will return these results to normal.

A key to determining treatment is distinguishing HHS from DKA; Table 1 compares the lab findings in these conditions.

Table 1



	HHS	DKA
Type of DM	2	1
Blood glucose, mg/dL	>600	>250
Serum osmolarity, mOsm/L	>320	>290
Serum ketones	Negative	Positive
Blood pH	>7.30	<7.30
Serum bicarbonate, mEq/L	>20	<15
Serum anion gap, mEq/L	8-12	>15

Knowledge Check

Question: What three key laboratory findings point to a diagnosis of hyperosmolar hyperglycemic state instead of diabetic ketoacidosis?

Answer: When HHS is suspected, a blood glucose level greater than 600 mg/dL, serum osmolarity greater than 320 mOsm/L, and the absence of ketoacids (no serum ketones, blood pH >7.3) confirms the diagnosis.

How Do We Manage Hyperosmolar Hyperglycemic State?

HHS is a medical emergency, so treatment focuses first on stabilizing the patient's condition. Initial steps include correcting volume depletion and hyperglycemia and stabilizing electrolytes.



Normal Saline

Typically, patients with HHS are typically massively volume depleted, often by 10 L or more. The first step to treatment of HHS is 0.9% saline (normal saline) given through an intravenous (IV) line at 1 L per hour to replenish volume. No more than 4 L should be given in the first 4 hours to avoid neurologic damage from massive fluid shifts. Large amounts of saline solution (>10 L) may be needed over the first few days. When serum glucose drops to less than 300 mg/dL, the IV saline should be switched to saline with 5% dextrose solution, to prevent overcorrecting the glucose level and triggering a hypoglycemic episode.

Insulin

Volume repletion alone can partially correct the hyperglycemia, but patients will still require insulin, often given through an IV line. Glucose levels should be measured every hour, as they progressively decrease from the effects of insulin.

Insulin should always be started after, not before, the initial normal saline infusion. This is to avoid massive fluid shifts that occur when insulin drives glucose and potassium into cells.

Potassium

Because of the rapid fall in serum potassium during treatment, potassium should be added to the IV solution when serum levels are <5.3 mEq/L. The patient will likely need both IV potassium initially and oral potassium later, because of the large urinary potassium losses from the osmotic diuresis. Do not give potassium if the serum potassium level is ≥ 5.3 mEq/L.

Although these treatments will stabilize the patient's emergency condition, the underlying trigger should be identified and treated as well. If the trigger was an infection or ischemic disease, treatments targeted at those conditions are necessary to prevent a recurrence of HHS. The patient should also be counseled on the importance of staying hydrated and maintaining good glycemic control. Educate the patient on how to recognize the early warning signs of HHS.

Knowledge Check

Question: What are the three basic steps in therapy for HHS?

Answer: The first step is IV 0.9% saline to correct volume depletion; insulin is then added, and potassium is given if the serum level is <5.3 mEq/L.



Case Connection

Thinking back to JV, what do you tell him and his daughter about the best way to treat his condition?

You explain that although insulin is a cornerstone of the treatment of HHS, you must give him IV fluids first. It is important to carefully monitor and slowly correct his fluids, blood sugar, and other abnormalities. If you give him insulin right away, his body won't be able to maintain the correct balance of fluids and electrolytes, which could cause problems in his heart and brain. You assure JV and his daughter that hospitalization is necessary; JV needs ongoing IV fluids and careful monitoring over several days to safely treat this condition.

Summary

- Hyperosmolar hyperglycemic state (HHS) is a medical emergency requiring hospitalization.
- In HHS, exceedingly high blood sugars are seen (often >1000 mg/dL) without significant ketosis.
- Triggers for HHS include medications, underlying infection, occult ischemia, and dehydration.
- HHS typically presents in older adult patients with type 2 diabetes mellitus (DM).
- Severe hyperglycemia and a hyperosmolar state lead to altered mental status and other manifestations and potential complications such as thrombosis and tachycardia.
- The pathophysiology of HHS is characterized by hyperglycemia and hyperosmolarity without significant ketosis; ketosis is prevented by the basal level of insulin in type 2 DM.
- In HHS, the symptoms of polyuria, polydipsia, and weight loss progress over several days, developing into severe symptoms that require hospital admission; neurologic symptoms such as seizures and confusion are common.
- Typically, a blood glucose level greater than 600 mg/dL, serum osmolarity greater than 320 mOsm/L, and the absence of ketoacidosis (no serum ketones, blood pH >7.3) will confirm the diagnosis of HHS.
- The lack of acidosis or elevated anion gap distinguishes HHS from DKA.
- The first step in treating HHS is providing intravenous 0.9% saline.
- Insulin is then started to reduce the serum glucose, and potassium repletion is usually required.
- Possible triggers for HHS should be identified and treated, and the patient must be counseled on staying hydrated and recognizing early warning signs.



Hyperosmolar Hyperglycemic State – Review Question

A patient has poorly controlled type 2 diabetes mellitus and presents to the emergency department with polyuria, polydipsia, blurred vision, and a serum glucose concentration of 780 mg/dL. Which of the following findings is most likely to be found on further evaluation?

- Decreased skin turgor
- Deep (Kussmaul) respirations
- Elevated jugular venous pressure
- Fruity breath odor
- Low blood pH

Diabetes & Other Disorders of the Endocrine Pancreas Part 6 of 8: Chronic Complications

Introduction

Case Connection

YW, a 59-year-old clerical assistant, was diagnosed with diabetes mellitus today when her hemoglobin A1c value was high. You arrange for diabetes education, order more labs, write prescriptions, and plan a follow-up visit in 2 weeks. YW is visibly upset and tells you, "I know diabetes is a death sentence. I watched my uncles fall apart piece by piece. One was blind and on dialysis for years, and the other had his foot amputated."

How will you counsel YW about preventing the vascular complications of diabetes mellitus? Consider your answer as you read, and we'll revisit YW at the end of the brick.

What Are the Chronic Complications of Diabetes Mellitus?

Some of the most dreaded chronic complications of diabetes mellitus (DM) develop slowly over years, usually requiring at least 5 years of poorly controlled disease to emerge. Such complications may lead to devastating outcomes such as blindness, renal failure requiring dialysis, and early death from heart attacks. The good news is that many can be prevented or slowed by control of blood glucose levels.

Because many of these complications are caused in part from injury to the blood vessels, they have traditionally been divided into two groups based on the size of the blood vessels involved: macrovascular and microvascular. Macrovascular complications of DM include those related to atherosclerosis, such as coronary artery disease (CAD), peripheral artery disease, and stroke. Microvascular complications of DM include diabetic retinopathy, diabetic kidney disease, and peripheral and autonomic neuropathy.

Other disease complications include arthropathy, cataracts, glaucoma, and more frequent infections. Here, we'll focus on macrovascular and microvascular complications. But first let's see how diabetes damages tissues in general.

How Does Diabetes Mellitus Damage Tissues?

DM is a complex, multisystem metabolic syndrome that causes pathologic changes throughout the body. Although hyperglycemia is the most visible sign, DM is in fact a

syndrome of insufficient insulin secretion, high glucagon formation, hyperlipidemia, and enhanced inflammation. Some patients develop complications even before a formal diagnosis of DM is made. In type 2 DM, insulin resistance is also a factor.

Hyperglycemic Injury in Diabetes Mellitus

Hyperglycemia itself plays a direct role in injuries to small blood vessels and proteins throughout the body. Some of the key biochemical mechanisms that contribute to chronic diabetic complications are shown in Figure 1.

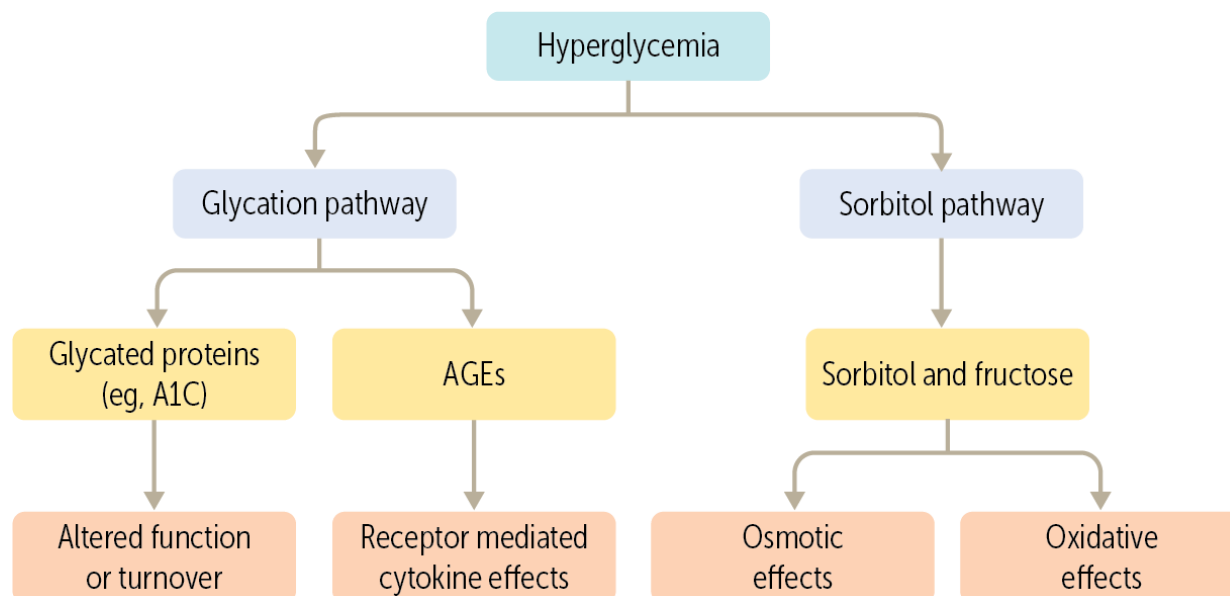


Figure 1. Image Credit: ScholarRx, used with permission.

Protein Glycation. The most important mechanism of hyperglycemic injury is the formation of glycoproteins, which occurs when regular body proteins are altered by high concentrations of glucose in the blood. Specifically, the high concentration of blood glucose enables **nonenzymatic glycation of proteins**, nucleotides, and lipids. In this process, a sugar molecule is attached onto a protein or other molecule, fundamentally changing its structure and impairing its function. These altered molecules are known as **advanced glycation end products (AGEs)**.

Clinical Correlation

An important clinical example of an advanced glycation end product is glycated hemoglobin. When hemoglobin binds to glucose from the blood, glycated hemoglobin (HbA1c) is formed; from the HbA1c value, we can get a general picture of the average blood glucose levels over the RBC's life cycle (approximately 120 days). This provides a stable measure of the patient's overall glucose control.

Depending on the tissue, AGEs may have several detrimental effects. AGEs increase vascular permeability, increase arterial stiffness, oxidize LDL cholesterol, and induce secretion of various cytokines. All this leads to **inflammation, altered vascular permeability, and tissue ischemia**.

Knowledge Check

Question: What is the effect of nonenzymatic glycation in DM?

Answer: *Nonenzymatic glycation of proteins alters their function and contributes to diabetic complications.*

Sorbitol. A less common mechanism of hyperglycemic injury is **sorbitol-mediated injury** that occurs in cells of the lens and retina and the peripheral nerves. These cells convert glucose to sorbitol but lack sorbitol dehydrogenase to subsequently convert sorbitol into fructose. So, sorbitol can build up in the cell, causing toxic osmotic and oxidative injury to tissues.

Inflammation in Diabetes Mellitus

DM is a highly inflammatory state. The activation of T cells with the production of cytokines contributes to many diabetic complications, especially **accelerated atherosclerosis** that causes CAD and strokes. Although the role of cytokines in microvascular complications of DM is less certain, interleukin 1 (IL-1), IL-6, IL-18, and tumor necrosis factor alpha (TNF- α) have been implicated in the development of diabetic kidney disease.

In small blood vessels, inflammation may lead to **platelet aggregation**, which decreases blood flow and can cause tissue ischemia. Inflammation is even implicated in the onset of type 2 DM, in which insulin resistance is partly caused by cytokines such as TNF- α and IL-6. Cytokine expression is one reason why obesity is linked to type 2 DM: obesity leads to high cytokine production by the adipocytes.



What Are the Macrovascular Complications of Diabetes Mellitus?

Macrovascular diabetic disease affects large arteries, causing accelerated atherosclerosis and high rates of CAD, peripheral arterial disease, and stroke. Hyperglycemia is only one factor that causes macrovascular disease; chronic inflammation, hypertension, hyperlipidemia, and high glucagon levels also contribute to injury to arteries that enables the development of atherosclerosis. The importance of nonhyperglycemic factors is supported by the observation that accelerated atherosclerosis in patients with diabetes often occurs in prediabetes, before any overt high blood glucose values are seen.

Coronary Artery Disease

The prevalence of CAD among patients with DM is remarkably high, especially in patients with poorly controlled disease. Because of the high incidence of CAD, the American Heart Association recommends that DM be considered a **CAD risk equivalent** instead of merely a risk factor. This means that the risk of having a major cardiovascular event (such as a stroke or myocardial infarction [MI]) in a patient with established diabetes without known CAD should be considered equal to the risk of developing such an event in a nondiabetic patient with known CAD. To say it another way, all patients with DM are treated as if they had CAD.

Knowledge Check

Question: What is the basis of CAD in patients with DM?

Answer: Atherosclerosis of the coronary arteries is the basis of CAD in patients with DM.

Clinical Presentation. Patients with DM and CAD may present with the typical symptoms of chest pain on exertion (angina) or at rest (unstable angina, MI). However, it's important to know that patients with DM may not experience the classic symptoms of MI, such as crushing chest pain. Symptoms may be nonspecific, such as nausea or dyspnea. This may be due to diabetic neuropathy that diminishes visceral sensation. Unfortunately, this leads to high rates of sudden arrhythmias and sudden death, because a patient's MI may go undetected in its early stages.

Diagnosis. Patients with DM who have even vague symptoms of chest pain or dyspnea or an abnormal resting ECG should receive exercise ECG testing or a similar test to screen for CAD.

Management. Screening for CAD in patients with DM without symptoms and with well-controlled glucose levels does not exclude future cardiac events. So the best approach is to address cardiovascular risk factors (hyperlipidemia, hypertension) and treat them in all patients with DM. As in all patients with diabetes, hemoglobin A1c should be less than 7%.



Preventive **low-dose aspirin** should be prescribed if patients have established coronary artery disease. The antiplatelet agent clopidogrel may be equally effective with fewer side effects.

Hyperlipidemia. To prevent CAD and other cardiovascular complications, statins are recommended for all patients with DM who are aged 40 to 70 years and older regardless of their LDL cholesterol level. In younger patients, statin therapy is initiated only for those with elevated LDL. In older patients, therapy is individualized.

Hypertension: The American Diabetic Association recommends that blood pressure should be controlled to less than 130/80 mm Hg in patients with DM and either known CAD or who are at high risk for cardiovascular disease. This may mean using multiple antihypertensive medications, beginning with an angiotensin-converting enzyme (ACE) inhibitor. In patients with diabetes with proteinuria, ACE inhibitors have a protective effect on the heart, kidneys, and even the eyes because they inhibit the hemodynamic and tissue effects of angiotensin II in patients with DM.

Knowledge Check

Question: What are the preferred drugs for treating hypertension in patients with DM?

Answer: ACE inhibitors are preferred for their protective effects on the heart and kidneys.

What Is Diabetic Retinopathy?

Diabetic retinopathy is the most common microvascular complication of DM. It affects up to 80% of patients who have had DM for at least 20 years and leads to around 10,000 new cases of blindness every year in the United States.

Pathophysiology of Diabetic Retinopathy

While the precise mechanism of retinopathy is unknown, factors that are thought to contribute include AGEs, which cause the basement membranes of the retinal vessels to thicken. Other factors include high growth hormone levels, platelet aggregation, and sorbitol weakening of capillary walls. High blood sugar levels cause retinal blood vessels to swell and leak, leading to **macular edema**. The blood vessels can also close, and these ischemic vessels prevent blood from reaching the macula; tiny clots of exudates ("floaters") form and can also affect vision. In advanced disease, new blood vessels can form in the retina (**neovascularization**). These vessels are fragile and often bleed into the vitreous, causing floaters and forming scar tissue that can ultimately lead to a detached retina.

Diagnosis of Diabetic Retinopathy

Dilation of the pupil followed by a comprehensive funduscopy exam of the retina allows the ophthalmologist to look for any hemorrhages, proliferation of vessels, or exudates (arrow, Figure 2). Although this can be done in a primary care office, DM guidelines suggest an annual visit to the ophthalmologist who will dilate the pupil; they have better equipment for spotting early signs of the disease.

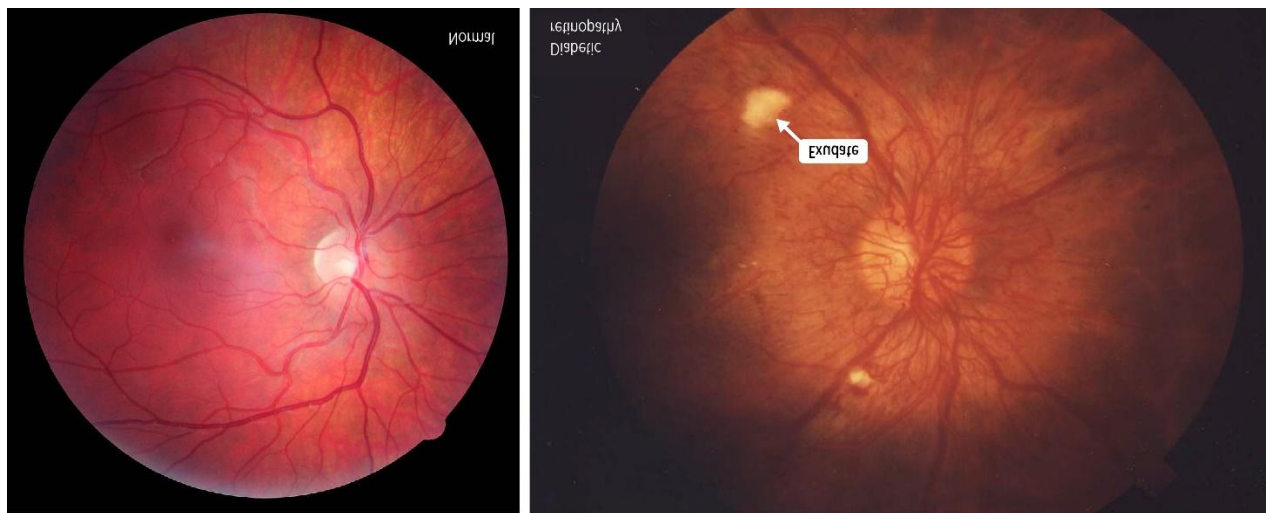


Figure 2. Image Credit: Left: Courtesy of Wikipedia Commons user Jan Kal@copy; from Brno, Czech Republic. Right: Courtesy of National Eye Institute, National Institutes of Health.

Knowledge Check

Question: How is retinopathy diagnosed?

Answer: In a fundoscopic exam, an ophthalmoscope is used to check the retina for the vascular changes that suggest retinopathy.

Management of Diabetic Retinopathy

Glycemic and blood pressure control are the best means of preventing this complication. Unlike in diabetic kidney disease, ACE inhibitors, although good for hypertension control, do not appear to have a separate protective effect. If it is already present, regular follow-up and good control of glycemia help slow progression. Laser photocoagulation delivered to the retina inhibits blood vessel proliferation and can preserve the patient's vision.

What Is Diabetic Kidney Disease?

Diabetic kidney disease is the most common cause of proteinuria, glomerular disease, and chronic kidney disease in the United States; it occurs in up to 25% of patients with



DM. Its onset is generally after 10 years of disease but can begin sooner if glycemic control is poor.

Clinical Presentation of Diabetic Kidney Disease

In the most familiar form of diabetic kidney disease, nephropathy progresses in an orderly, relentless way. It starts asymptotically, with very small (but abnormal) amounts of albumin in the urine (50-300 mg of albumin in the urine/24 hours), to heavier albuminuria (>300 mg/24 hours), to nephrotic-range proteinuria (>3500 mg/24 hours), to progressive decline in the glomerular filtration rate (GFR), then end-stage renal disease (ESRD). As with any glomerular disease, advanced nephropathy causes symptoms such as edema, fatigue, and nausea as renal function declines.

However, it is now known that some diabetic patients may have decreased GFR even before the urine albumin increases. As many as half of patients with type 2 diabetes and decreased GFR have no albuminuria. These patients may never develop nephrotic syndrome but can have progressive loss of GFR and develop ESRD. This group seems to have more renal interstitial disease with scarring. Some now feel that in determining prognosis, the decline in GFR is as or more important than increasing albuminuria.

Pathophysiology of Diabetic Kidney Disease

Diabetic kidney disease is most typically a glomerular disease that leads to expansion of the mesangial matrix and thickening of the glomerular basement membrane. Albumin leaks into the urine through the damaged membrane. The glomerular damage is thought to occur by a combination of AGE formation, high intraglomerular pressure, and cytokine release. Eventually, increased glomerular capillary pressure and hemodynamic damage cause glomerular sclerosis and loss of kidney function. A form of sclerosis called nodular sclerosis is shown in Figure 3. Other patients have more severe interstitial fibrosis and scarring, although the glomeruli are more preserved.

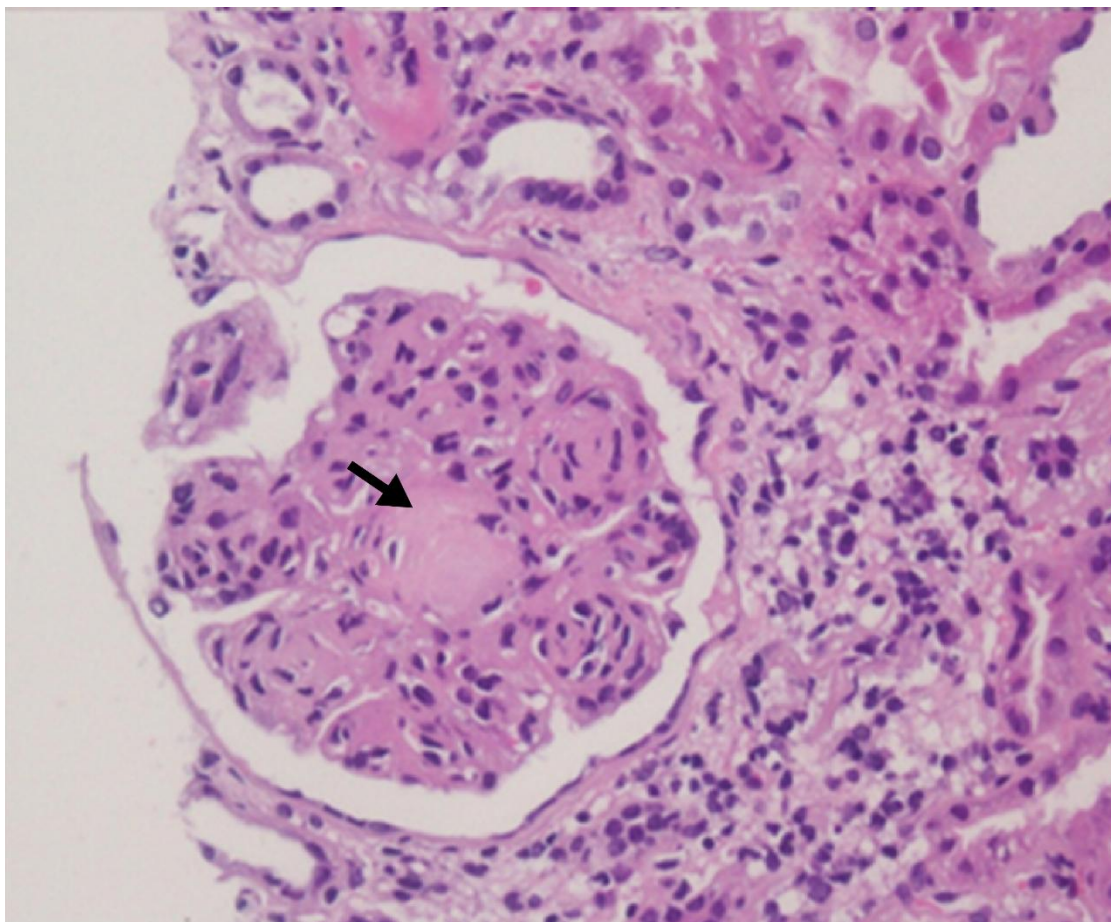


Figure 3. Image Credit: Courtesy of Satirapoj B, Tassanasorn S, Charoenpitakchai M, et al. journals plos.org. 2015;10(4):e0124055. doi:10.1371/journal.pone.0124055.

Diagnosis of Diabetic Kidney Disease

The key is to detect this disease in its earliest stage with low amounts of urine albumin and preserved GFR. In later stages, we can slow progression of the disease down but not prevent it.

The way to detect nephropathy is with an **annual test for urine albumin**. This is done most simply by using a spot microalbumin test, which uses the ratio of urine albumin to urine creatinine as a good estimate for the 24-hour amount. A level of 30 to 300 mg albumin per 24 hours indicates abnormally increased urine albumin, which can often be reversed with ACE inhibitor treatment. The GFR should be estimated annually with measurement of the serum creatinine and calculation of the estimated GFR (eGFR).

Knowledge Check

Question: What is the best way to screen for nephropathy?

Answer: An annual test for urine albumin is the best way to screen for nephropathy.

Management of Diabetic Kidney Disease

Strict blood pressure control (<130/80 mm Hg) using ACE inhibitors can often reverse and prevent kidney disease if started early, when albuminuria is mild. End-stage disease is treated with transplantation or dialysis.

What Is Diabetic Neuropathy?

DM affects the nervous system in two ways, via diabetic peripheral neuropathy and diabetic autonomic neuropathy.

Peripheral Neuropathy

Peripheral neuropathy is the presence of symptoms or signs of peripheral nerve dysfunction in patients with DM after the exclusion of other causes. Nerve damage may present as sensory-motor polyneuropathy, pure sensory neuropathy, or mononeuropathy.

Clinical Presentation. Symptoms commonly include foot pain, tingling, and numbness (paresthesias), often in a “stocking-glove” distribution. The severity of these complications depends on how well blood glucose is controlled and how long DM has been present. Diabetic neuropathy starts early during prediabetes. Sensory and motor loss puts the patient at risk for foot ulcers that can become infected, especially when combined with accelerated atherosclerosis, which narrows arteries and causes peripheral vascular disease (Figure 4).



Figure 4. Image Credit: Courtesy of Wikimedia Commons user Intermedichbo.

Patients may also develop **diabetic arthropathy** presenting with warmth and redness in the foot or ankle. It's probably related to the lack of proprioception, subjecting the joints to additional strain and trauma, but may also be related to bone abnormalities. It is initially treated with casting, then orthopedic footwear.

Pathophysiology. The precise cause of neuropathy is not understood. Sorbitol and AGE damage of myelinated nerves and Schwann cells is likely. Oxidative injury to the small arteries supplying the nerves may also contribute. So, although diabetic neuropathy is usually considered a “microvascular complication,” direct injury to nerves may be more important.



Diagnosis. Sensation should be checked annually with a vibratory sense test, a pinprick test, and a monofilament, which is very sensitive in detecting subtle sensory deficits.

Foot ulcers are the most likely consequence of peripheral neuropathy. On each clinic visit, check for any skin changes, muscle and ligament changes, poor wound healing, ulcers, and fungal infections of the toenails or between the toes.

Knowledge Check

Question: What is the most common complication of peripheral neuropathy?

Answer: Foot ulcers are the most common complication of peripheral neuropathy.

Management. To avoid foot ulcers, patients should be taught to do weekly foot exams at home, wear comfortable footwear, and avoid walking with bare feet. A variety of pharmacologic agents are available to manage the pain associated with diabetic neuropathy, including pregabalin, duloxetine, and gabapentin as first-line agents. Fungal toe infections should be treated with appropriate antifungal drugs.

Autonomic Neuropathy

This form of diabetic neuropathy leads to autonomic dysfunction of both the sympathetic and parasympathetic systems, causing a range of potentially disabling symptoms.

Clinical Presentation. Patients may have a range of autonomic nervous system disorders, including the following:

- Orthostatic hypotension, which can cause falls and fractures
- Resting tachycardia, exercise intolerance
- Decreased gastrointestinal (GI) motility (gastroparesis, nausea, constipation), which can lead to anorexia, malnutrition, and weight loss
- Increased GI motility (diarrhea)
- Bladder dysfunction (incontinence or urinary retention)
- Erectile dysfunction
- Decreased sweating of the hands and feet, leading to skin dryness

Pathophysiology. The causes are multifactorial and similar to those of peripheral neuropathy: AGE, inflammation, sorbitol, and microvascular damage.

Orthostatic hypotension is multifactorial in origin. The most obvious cause is abnormal sympathetic regulation of smooth muscle contraction in blood vessels. Also, there is decreased sympathetic input to the renal juxtaglomerular (JG) cells, which make renin. Less renin means less angiotensin II and aldosterone, part of the renin-angiotensin-aldosterone system of blood pressure and blood volume control. The resulting low



aldosterone makes this one of the most common forms of **hypoaldosteronism**, which can in turn lead to volume depletion and hyperkalemia.

Diagnosis. If patients have symptoms of autonomic neuropathy, it's important to rule out other reversible causes. For example, patients with hypotension need an ECG and cardiac evaluation to look for heart failure, those with GI symptoms may need endoscopy to look for ulcers, and those with bladder dysfunction may benefit from cystoscopy.

To diagnose gastroparesis, a gastric emptying study is the standard. It uses an ingested radioisotope to follow the progress of a meal through the stomach and upper GI tract.

Management. For orthostatic hypotension, options include compression stockings and physical therapy including resistance exercises. Drugs that increase vascular tone, such as midodrine, may be tried for severely symptomatic patients. Patients with hypoaldosteronism may benefit from fludrocortisone.

Diabetic gastroparesis due to autonomic neuropathy is difficult to treat. Medications that may be useful for severe nausea include metoclopramide, diphenhydramine, and ondansetron, but all have side effects that may limit their use.

Table 1 summarizes current ADA guidelines for prevention and screening for chronic diabetic complications.

Table 1

Disorder	Test	Frequency or target
CAD	Exercise stress test (or equivalent)	If symptomatic, history of stroke or peripheral artery disease, or ECG changes.
Hyperlipidemia	Cholesterol panel	Annually if taking statin or ≥ 40 years old; every 5 years if normal lipid panel and < 40 years old.
Hypertension	Monitor at home and in office 4×/year	140/90 mm Hg or 130/80 mm Hg if known cardiovascular disease.
Retinopathy	Dilated comprehensive eye exam by ophthalmologist or optometrist	Yearly



Disorder	Test	Frequency or target
Nephropathy	Urine microalbumin:creatinine ratio	Yearly
Peripheral neuropathy	Monofilament foot exam and vibration and temperature testing	Yearly

Case Connection

Thinking back to YW, how do you counsel her about preventing vascular complications of DM?

You explain to YW that the problems her uncles had were caused by high blood sugar causing damage to small blood vessels, which can affect the kidneys, eyes, nerves, and peripheral circulation. However, keeping blood sugar in control can significantly reduce the risk of developing these complications. You assure her that you will work together and that with lifestyle measures and medication, she can keep her blood sugar in a healthy range and avoid such complications.

Summary

- Complications are partly caused by macroangiopathy and microangiopathy.
- The primary mechanism causing vascular complications in diabetes mellitus (DM) is vascular damage from advanced glycation end products (AGEs).
- Macrovascular complications of DM are coronary artery disease (CAD), peripheral artery disease, and stroke.
- Symptomatic patients need tight blood pressure control and an exercise stress test to screen for CAD.
- The best ways to prevent CAD in patients with DM is early diagnosis and addressing and treating the risk factors.
- Factors that are thought to contribute to macrovascular complications include AGEs, high levels of growth hormone, platelet aggregation, and sorbitol weakening of capillary walls by sorbitol.
- Angiotensin-converting enzyme (ACE) inhibitors, statins, antihypertensives, and aspirin are all used for prevention of complications.
- Screening for retinopathy is done with a dilated fundus exam.
- Glycemic and blood pressure control are preventative as well as help slow progression if retinopathy is already present; laser photocoagulation can preserve the patient's vision in advanced disease.



- Screening for nephropathy is done with annual urine albumin tests.
- ACE inhibitors can control blood pressure and prevent kidney disease if used when albumin is 30 to 300 mg/24 hours; they can slow progression in later stages.
- Screening for peripheral neuropathy is done with foot exams and sensation tests.
- To manage the pain associated with diabetic neuropathy, first-line agents include pregabalin, duloxetine, and gabapentin.
- Autonomic neuropathy leads to autonomic dysfunction of both the sympathetic and parasympathetic systems, causing a range of potentially disabling symptoms.

Chronic Complications - Review Question

Which is the most common microvascular complication of diabetes mellitus?

- Coronary artery disease
- Diabetic ketoacidosis
- Diabetic kidney disease
- Peripheral neuropathy
- Retinopathy

Diabetes & Other Disorders of the Endocrine Pancreas Part 7 of 8: Hypoglycemia

Introduction

Case Connection

ZM, a 60-year-old female who takes glyburide for diabetes mellitus (DM), presents to your clinic on Monday morning with a racing heart, sweating, and dizziness. The medical assistant reports a critically low blood glucose result. After ZM has some apple juice and graham crackers, she tells you that this has happened twice in the past week. The first time, she ate something and felt better. Three days ago, she had similar symptoms, prompting her husband to take her to the hospital. She had a very low blood glucose level and was admitted for observation, but she had no further hypoglycemia in the hospital and was discharged yesterday. She and her husband are very worried.

What could be causing ZM's episodes of hypoglycemia? Consider your answer as you read, and we'll revisit ZM at the end of the brick.

What Is Hypoglycemia?

Hypoglycemia, commonly known as "low blood sugar," is a glucose level below 70 mg/dL. The risks of hypoglycemia (and insulin administration) and why it needs to be corrected promptly are key points of our discussion. We'll start by reviewing how the body maintains a normal serum glucose level and then discuss the causes and management of hypoglycemia.

Fasting vs Reactive Hypoglycemia

It's useful to categorize patients with hypoglycemia into fasting vs reactive causes. Fasting hypoglycemia occurs in the morning before eating or occasionally during the day if a meal is missed. In contrast, reactive hypoglycemia is only seen 2 to 4 hours after a meal.

Fasting hypoglycemia is often severe and symptomatic; it is typically caused by side effects from medication (especially glucose-lowering drugs for diabetes mellitus [DM]) or a medical problem such as low cortisol or chronic kidney or liver disease. The underlying problem needs to be identified and corrected immediately to avoid serious problems.

Reactive hypoglycemia is a diagnosis popularized in recent decades when patients complained of fatigue, confusion, or jitteriness after meals. Unlike fasting hypoglycemia,



it seldom causes glucose levels to drop low enough to induce severe symptoms. It's seen most often in women aged 25 to 35 years, especially those who are overweight, and it may sometimes be a precursor to type 2 DM. It can occur after upper gastrointestinal (GI) surgery (eg, bariatric surgery or gastrectomy). Such surgeries can cause rapid glucose transit and absorption in the small intestine, leading to excessive insulin increase after a meal and hypoglycemia (dumping syndrome). The prognosis is good in most cases of reactive hypoglycemia, and it usually goes away on its own or after dietary modifications.

Patients who have reactive hypoglycemic symptoms without documented decreases in blood glucose are diagnosed with **idiopathic postprandial syndrome** (sometimes called pseudohypoglycemia because of the absence of actual hypoglycemia). The cause is unknown, but symptoms are often relieved with frequent snacks and avoidance of simple sugars in the diet. Although this is a rarely diagnosed condition throughout most of the world, it is widely diagnosed in United States, prompting the American Diabetes Association and the Endocrine Society to issue statements that this condition is probably overdiagnosed.

How Do Patients With Hypoglycemia Present?

The blood glucose level at which a given patient may develop hypoglycemic symptoms varies. In general, when the **blood glucose levels are less than 50 to 55 mg/dL**, patients will show hunger, activation of the sympathetic and parasympathetic autonomic nervous systems, and impaired cognitive function. Commonly, older adult patients experience fewer of these symptoms, which makes it harder to make the initial diagnosis.

The classic hypoglycemic symptoms include fight-or-flight findings, such as:

- Tachycardia
- Sweatiness
- Tremor
- Dizziness, progressing to syncope (passing out)

These sympathetic nervous system effects result from the action of adrenaline (epinephrine), insulin's counterregulatory hormone that's released from the adrenal gland when blood glucose decreases.

Altered brain function includes:

- Confusion progressing to coma
- Irritability
- Headache

The cognitive impairment happens because the brain cannot itself make glucose or store glycogen, so it totally relies on circulating glucose for its normal function. This makes the brain vulnerable to hypoglycemia.

Knowledge Check

Question: Why is the brain more susceptible to hypoglycemia than other organs?

Answer: The brain cannot itself store glycogen or can make glucose, so it relies on circulating glucose for its normal function. Hence, it is vulnerable to hypoglycemia

What Is the Pathophysiology of Fasting Hypoglycemia?

Most of the hypoglycemia we see in medical practice is caused by fasting hypoglycemia and requires a good medical history with review of medications.

Let's now look at how hypoglycemia can arise, including high insulin levels, reduced hepatic glucose formation, and low cortisol states.

High Exogenous Insulin Levels in Patients with Diabetes Mellitus

Because insulin lowers blood glucose, it's not surprising that several of the most common causes of hypoglycemia are caused by high insulin levels. Far and away, most cases of hypoglycemia are caused by drugs that treat diabetes by raising insulin levels.

Insulin Injections. Hypoglycemia is common in patients with type 1 or 2 DM who inject insulin. Glucose needs tight control, but it is often hard for patients to keep their blood glucose at good target levels without overshooting and developing hypoglycemia. This is especially common if they miss meals or lose their appetite (reducing caloric intake) or if they take too much insulin.

Oral Antihyperglycemic Drugs. These are used to treat hyperglycemia in type 2 DM. Several of them work by increasing pancreatic insulin secretion, and these are the ones that most often cause hypoglycemia. The **sulfonylureas**, such as glyburide and glipizide, and the meglitinides, such as repaglinide, are commonly implicated.

Clinical Correlation

Sometimes, patients use drugs not prescribed by their clinician. This includes the use of insulin or antihyperglycemic agents seen in some patients with psychiatric disorders who self-administer to gain attention or sympathy or to attempt suicide. Healthcare professionals with malingering or factitious disorder are also at higher risk of self-administering these medications because of their easy access to them.



High Endogenous Insulin Levels in Patients Without Diabetes Mellitus

Insulin levels can also increase in nondiabetic disease states, and these conditions must be ruled out prior to attributing the increase in insulin to use of injections or oral antihyperglycemic drugs. Some conditions that may present with high insulin levels include:

- **Chronic kidney disease (CKD)** may lead to hypoglycemia in patients with and without DM. Insulin is renally cleared, so decreased renal filtration raises insulin levels. Patients with CKD and DM may have decreased clearance of both prescribed insulin and of oral antihyperglycemic drugs. In addition, the kidney is a major source of gluconeogenesis, which decreases in CKD. Finally, patients with CKD may become anorexic, decreasing caloric intake.
- **GI surgery**, such as gastrectomy or bariatric surgery, may cause rapid GI glucose transit and absorption in the small intestine, leading to an excessive insulin response after a meal.
- **Nondiabetic drugs** can potentially cause or worsen hypoglycemia by increasing insulin release. These include aspirin overdose, pentamidine, quinolone antibiotics, and monoamine oxidase inhibitors.
- **Insulinomas** are rare tumors of the pancreatic islet β cells that release large amounts of insulin.
- **Congenital hyperinsulinism**, though rare, is the most common cause of hypoglycemia in infants and children. It may be caused by abnormalities in the sensing of glucose by islet β cells.

Knowledge Check

Question: What drugs most often increase pancreatic insulin release?

Answer: Antihyperglycemic drugs, such as glyburide, increase insulin levels; less commonly, drugs such as aspirin and quinolone antibiotics can increase insulin levels.

Decreased Hepatic Gluconeogenesis and Glycogenolysis

The body maintains a normal serum glucose level when energy demands are high (such as while exercising) by two processes: gluconeogenesis and glycogenolysis. Both occur mostly in the liver. Gluconeogenesis converts nonglucose substrates, such as amino acids, lactate, and fructose, into glucose. Glycogenolysis converts stored glycogen into glucose. Some causes of hypoglycemia interfere with these processes, causing the serum glucose to drop, especially in times of stress.



Liver Disease. Any liver disease can cause hypoglycemia because the liver is largely responsible for gluconeogenesis and glycogenolysis. This is especially common in alcoholic cirrhosis.

Alcohol Use. Alcohol also inhibits hepatic gluconeogenesis, sometimes causing hypoglycemia. Ethanol is mostly metabolized by the liver. This metabolism occurs in two steps, the first of which is catalyzed by the enzyme alcohol dehydrogenase and the second of which is catalyzed by the enzyme aldehyde dehydrogenase (Figure 1).

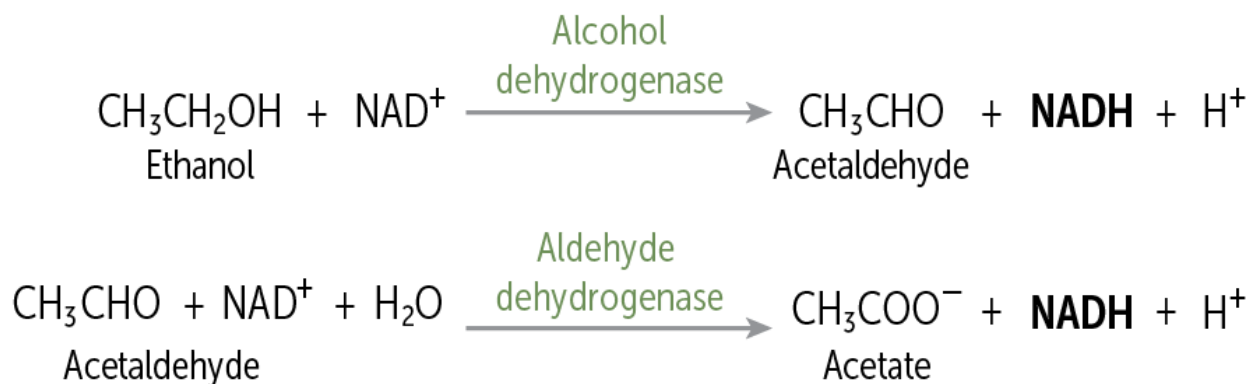


Figure 1. Image Credit: ©ScholarRx, used with permission.

So you can see that ethanol consumption and metabolism lead to an accumulation of reduced nicotinic adenine dinucleotide (NADH). This high concentration of NADH inhibits gluconeogenesis by preventing the oxidation of lactate to pyruvate.

Knowledge Check

Question: How does alcohol consumption cause hypoglycemia?

Answer: Metabolism of ethanol increases NADH, which inhibits gluconeogenesis and causes hypoglycemia.

Congenital Enzyme Deficiencies. Several rare congenital deficiencies of liver enzymes can impede gluconeogenesis. In addition, ketotic hypoglycemia of childhood is a poorly understood disease that leads to hypoglycemia with high ketone levels. It may involve a limited rate of hepatic glucose production.

Starvation. Starvation and anorexia nervosa cause profound malnutrition and decrease glycogen stores, limiting the ability to increase the blood glucose with glycogenolysis when needed.

Extreme Exercise. Marathon running, for example, depletes glycogen stores, leading eventually to the need to burn fatty acids for energy instead; this is the familiar drop in energy known as “hitting the wall” or “bonking.” Sometimes overt hypoglycemia can occur, as stressed cells avidly take up glucose and new glucose cannot be made quickly enough to meet the body’s needs. This is especially the case in athletes with DM, who

may need to increase carbohydrate intake and reduce their insulin dose before and after heavy exercise.

Other Causes

Cortisone, thyroid hormone, growth hormone (GH), and glucagon are hormones that all act to increase glucose levels, so deficiencies of these hormones can lead to hypoglycemia. **Low-cortisol** states (eg, adrenal insufficiency), can result from these deficiencies and often present with hypoglycemia.

Systemic **infections** increase the risk of hypoglycemia by an unclear mechanism, but it may be related to reduced gluconeogenesis or increased peripheral glucose uptake and utilization by infection-stressed cells.

Pregnancy is mostly associated with hyperglycemia. However, it also increases the risk of hypoglycemia in women with DM and is likely caused by diet or medication changes. This is most common in women younger than 25 years.

Knowledge Check

*Question: Most common causes of hypoglycemia: **SHAKIER***

*Answer: Most common causes of hypoglycemia: **SHAKIER**:*

*Sulfonylurea drugs, **Hepatic failure**, **Alcohol**, **Kidney failure**, **Insulin/Infection**, **Endocrine disorders**, **Reactive hypoglycemia***

What Is the Pathophysiology of Reactive Hypoglycemia?

Reactive hypoglycemia occurs when patients have hypoglycemia, fatigue, confusion, or jitteriness after meals. Many more people have these symptoms without actual hypoglycemia, so true reactive hypoglycemia is uncommon. It occurs most often after GI surgical procedures.

Postoperative

Hypoglycemia after meals is most often caused by upper GI surgical procedures.

Alimentary Hypoglycemia. Hypoglycemic episodes after stomach surgery (gastrectomy, gastroenterostomy, or vagotomy and pyloroplasty) may occur, with patients having sympathetic symptoms (tachycardia, sweatiness, tremor) 3 to 4 hours after meals.

Postbariatric Surgery. Some patients who have undergone Roux-en-Y gastric bypass as treatment for morbid obesity have experienced reactive (postprandial) hypoglycemia.

How Do We Diagnose Hypoglycemia?

The key to diagnosing hypoglycemia is the Whipple triad: document low blood glucose at the same time as characteristic symptoms and then show that both are corrected with glucose administration (Figure 2).

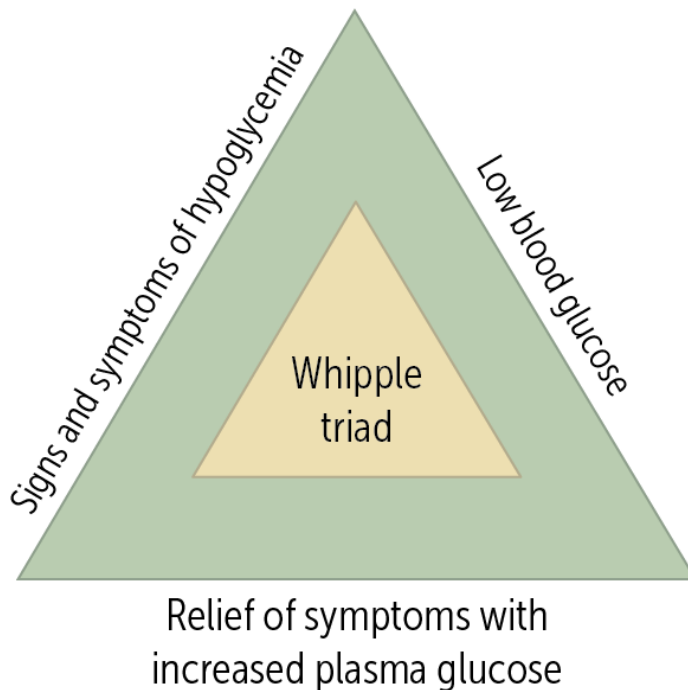


Figure 2. Image Credit: ©ScholarRx, used with permission.

Documenting **blood glucose levels below 55 mg/dL during symptoms** is the first step.

Knowledge Check

Question: How is the Whipple triad used to diagnose hypoglycemia?

Answer: To be diagnosed with hypoglycemia, patients should have a low blood-glucose reading in the presence of characteristic symptoms, and both should be corrected after treatment with glucose.

Confirmatory Tests

Sometimes, the diagnosis is not clear after initial testing. When that's the case, you can use additional stimulation tests to sort things out:

- **Fasting hypoglycemia:** A supervised 72-hour fast is performed, checking blood glucose and insulin levels every 6 hours until symptoms occur or the blood glucose drops below 55 mg/dL.
- **Reactive hypoglycemia:** After giving glucose either in a mixed meal or as simple glucose, check glucose levels hourly for 5 hours. A positive test shows



characteristic symptoms with a blood glucose below 55 mg/dL. This test has been criticized as poorly reproducible.

Testing for Chronic Medical Diseases

Because fasting hypoglycemia is often caused by underlying medical diseases, other blood tests are used to identify them:

- **Liver tests:** Normal levels of aspartate transaminase, alanine transferase, alkaline phosphatase, bilirubin, and albumin help rule out hepatic disease.
- **Cortisol:** Check for a low morning cortisol, and do an adrenocorticotrophic hormone stimulation test if hypocortisolism (ie, Addison disease) is suspected.
- **Thyroid hormone:** Check the level of thyroid-stimulating hormone to rule out hypothyroidism.
- **Growth hormone:** GH deficiency can be ruled out by measuring levels of GH and insulin-like growth factor 1 (IGF-1).

Tests for Insulinoma or Drug Ingestion

If no underlying medical disease is evident, other serum tests can help identify unusual causes of fasting hypoglycemia, such as factitious drug use or insulinoma. The tests are best done under medical supervision when the blood glucose level is low:

- **Insulin level:** This will be elevated in both exogenous and endogenous insulin excess. Before launching an expensive evaluation into insulinoma or factitious insulin use, ensure that other common conditions that increase insulin levels, such as drugs and CKD, are excluded.
- **Proinsulin level:** Proinsulin is the precursor molecule to endogenous insulin; it is not present in injected insulin. The level is usually less than 20% of total insulin in normal patients. It's often more than 70% in patients with insulinomas.
- **C-peptide level:** C-peptide is part of precursor insulin (proinsulin) that is cleaved off during the normal processing of endogenous insulin, so it's not present in injected insulin. It's elevated in cases of high endogenous insulin secretion.
- **Sulfonylurea level:** This is used to detect factitious use of antihyperglycemic drugs.

Clinical Correlation

The C-peptide test is more often done to help distinguish type 1 DM (no C-peptide because of no insulin secretion) from type 2 DM (high insulin levels in many cases).

Table 1 summarizes use of these tests.



Table 1

Condition	Insulin	Proinsulin	C-Peptide	Sulfonylurea
Factitious insulin use	↑	Normal to ↓	Normal to ↓	0
Factitious sulfonylurea use	Normal to ↑	Normal to ↑	↑	↑
Insulinoma	↑↑	↑↑	↑	0

Knowledge Check

Question: What would be the proinsulin and C-peptide levels after injecting insulin into a patient who does not have diabetes?

Answer: Because pharmacologic insulin is preprocessed, it contains no proinsulin or C-peptide, so levels would be normal.

How Do We Manage Hypoglycemia?

Most important is the acute, immediate correction of low blood glucose levels in any symptomatic patients to avoid irreversible neurologic damage. Patients with altered mental states may require an intravenous (IV) dextrose solution. Injected glucagon can also be given if IV access is unavailable. Patients who are alert can be given fruit juice, soda, milk, honey, or candy.

Fasting Hypoglycemia

For management of chronic fasting hypoglycemia, first adjust medications (including insulin and antihyperglycemic drugs in DM), and treat underlying renal, hepatic, or endocrine disease. Medical options include inhibitors of insulin secretion, such as octreotide and diazoxide.

Clinical Correlation

Octreotide is also used to reduce GH secretion in acromegaly, as a vasoconstrictor in patients with esophageal bleeding, and in patients with severe chronic liver disease who have concurrent renal failure (hepatorenal syndrome).

Reactive Hypoglycemia

For reactive hypoglycemia, simple dietary changes can often help. Try frequent meals and snacks, limit intake of refined sugar, and increase physical activity.



For resistant reactive hypoglycemia, α -glucosidase inhibitors, such as acarbose and miglitol, may help. These medications inhibit pancreatic enzymes, leading to delayed GI glucose absorption. This slowed absorption reduces postprandial hyperglycemia, which might otherwise cause a spike in insulin, eventually leading to symptomatic hypoglycemia.

Clinical Correlation

Acarbose is more often used as an oral antihyperglycemic agent for type 2 DM to lower serum glucose levels. The slowed metabolism of carbohydrates in the gut lowers glucose absorption, and this, in turn, lowers glucose levels.

Case Connection

Thinking back to ZM, what could be causing her episodes of hypoglycemia? To avoid unnecessary medical expense, you obtain lab records from her hospitalization and find that her kidney, liver, insulin, and thyroid tests were all normal. You ask if her diet or medications have changed, and she says, "Not in general. I did take an extra dose of my glyburide just one time. About a week ago, my finger-stick glucose result was high; I felt like I should do something." Glyburide is a sulfonylurea drug, and hypoglycemia is an important adverse effect. You counsel ZM and her husband about the importance of sticking to the doses prescribed and calling her clinician if she is worried about a finger-stick result.

Summary

- Hypoglycemia, or low blood glucose, is commonly caused by high insulin levels or by impaired hepatic gluconeogenesis or glycogenolysis.
- The most common cause of hypoglycemia is excessive insulin or sulfonylurea drugs in patients with diabetes mellitus.
- Common symptoms include dizziness, tremors, tachycardia, weakness, sweatiness, and confusion.
- Fasting hypoglycemia presents with symptoms on getting up in the morning.
- Reactive hypoglycemia presents with symptoms 2 to 4 hours after meals.
- Diagnosis includes documenting a blood glucose level under 55 mg/dL in the presence of symptoms and seeing correction of both after glucose is administered.
- Important secondary causes such as liver disease, low cortisol, hypothyroidism, and chronic kidney disease should be ruled out as part of the work-up.



- The levels of insulin, proinsulin, and C-peptide are useful in distinguishing endogenous from exogenous insulin excess.
- Treatment of acute hypoglycemia is immediate administration of intravenous or oral glucose.
- Chronic management includes dietary and medication adjustments, treating underlying causes and, if needed, medications such as octreotide, diazoxide, and acarbose.

Hypoglycemia - Review Question

Which of the following would most likely cause hypoglycemia?

- Decreased gluconeogenesis
- Decreased insulin
- Glucagon-secreting tumors
- Increased cortisol
- Increased glycogenolysis

Diabetes & Other Disorders of the Endocrine Pancreas Part 8 of 8: Pancreatic Tumors

Introduction

Case Connection

SG is a 44-year-old female with poorly controlled diabetes who presents with a painful, swollen right leg, constipation, and frequent urination. Using Doppler ultrasonography, you discover a thrombus in her popliteal vein. SG is distraught. “I’m doing everything right, and everything is still going wrong. I’m losing weight, and I feel like I am falling apart!” On labs, her serum calcium concentration is high, and you are concerned about an underlying problem involving several endocrine abnormalities. You check her glucagon level: it is greatly elevated. On ultrasound, you see a poorly defined mass in the pancreatic body. You tell SG that you think she has a tumor in her pancreas called a glucagonoma. “Can it be removed?” she asks. What will you tell SG about the first step in her treatment? Consider your answer as you read, and we’ll revisit SG at the end of the brick.

What Are Endocrine Pancreas Neoplasms?

Endocrine neoplasms (tumors) of the pancreas arise from pancreatic islet cells—called the islets of Langerhans—that normally produce and release hormones into the bloodstream (Figure 1).

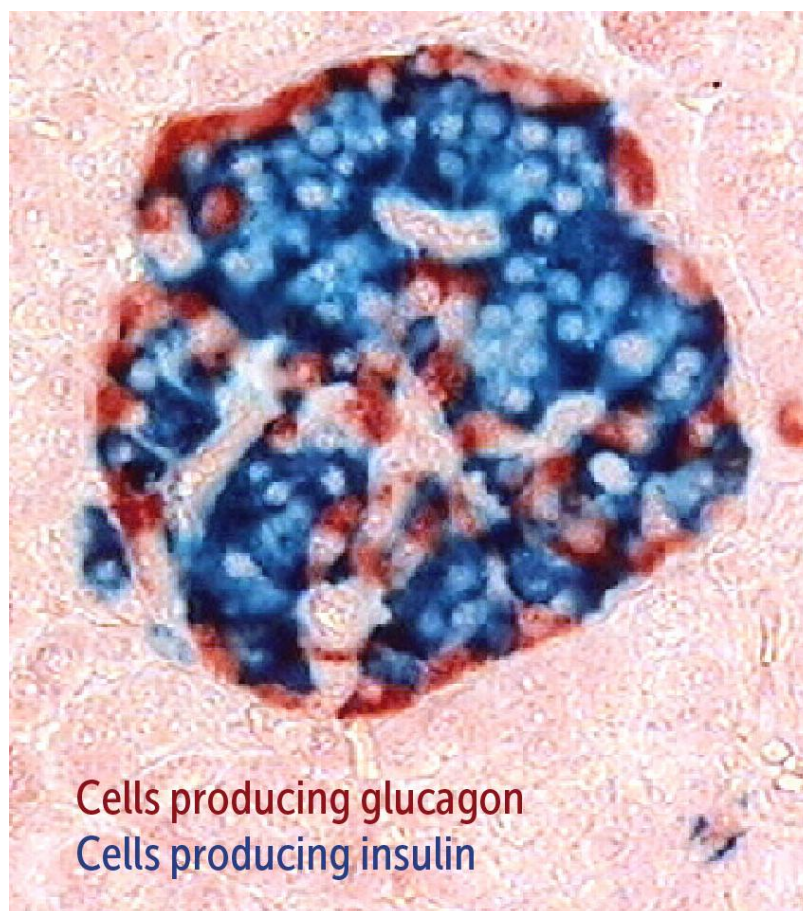


Figure 1. Image Credit: Courtesy of Wikipedia Commons user Afferent.

These tumors are variously called **islet cell tumors** and **neuroendocrine tumors**, because they arise from primitive neuroendocrine cells, similar to small cell cancer of the lung and well differentiated ("carcinoid") tumors of the small intestine and lung. As neuroendocrine tumors, they may produce endocrine hormones.

These tumors can be functional (75% of tumors) or nonfunctional (25%). Functional tumors are named after the hormones they produce; for example, insulinomas secrete insulin. The symptoms of these tumors mimic those of other disorders, so they are often difficult to diagnose. For this reason, patients commonly present with advanced or metastatic disease.

Here we'll look at some of the most common types of these rare tumors, going from most to least common: insulinomas, VIPomas, glucagonomas, and somatostatinomas.

Clinical Correlation

The more common **exocrine tumors** of the pancreas, commonly called “pancreatic cancer,” arise from the portions of the pancreas that are involved in the secretion of digestive products. The most common example is pancreatic adenocarcinoma.

What Are Insulinomas?

Although insulinomas are the most common pancreatic endocrine tumor, they are still rare, with an estimated incidence of 3 to 4 per 1,000,000 people.

Clinical Presentation of Insulinomas

Insulinomas secrete insulin at levels too high to be responsive to negative feedback from counterregulatory hormones. As a result, patients present with **hypoglycemia**, particularly during periods of fasting. Common symptoms of hypoglycemia are confusion, headache, disorientation, and even coma in extreme cases. The hypoglycemia then triggers a release of catecholamines, which cause tremors, sweating, and palpitations.

Although insulinoma is a rare cause of hypoglycemia (insulin and antihyperglycemic drugs for diabetes mellitus (DM) are by far the most common causes), it should be considered in any patient with recurrent hypoglycemia who is not taking medications for DM. Before making a diagnosis, it is important to first rule out other causes of hypoglycemia, such as DM treatments, severe liver disease, poor nutrition, and gastric bypass surgery.

Pathophysiology of Insulinomas

Insulinomas arise from β cells of the pancreas, which secrete insulin. In insulinomas, insulin release is not subject to regulation, leading to high insulin levels. The glucose decreases, and catecholamines and cortisol (counterregulatory hormones) increase.

Although most insulinomas occur in isolation, some patients have them as part of **multiple endocrine neoplasia type 1 (MEN1) syndrome**, in which also they might also have parathyroid adenomas (causing high parathyroid hormone and hypercalcemia), gastrinomas (causing peptic ulcers), and/or pituitary tumors (especially prolactinomas). These patients have autosomal-dominant abnormalities of the MEN1 gene. This gene is thought to function as a tumor-suppressor gene.

Diagnosis and Treatment of Insulinomas

The hallmark of an insulinoma is the failure of endogenous insulin secretion to be suppressed during hypoglycemia. The Whipple triad of symptoms is used in the clinical diagnosis of insulinoma (Figure 2):

- Symptoms of hypoglycemia
- Plasma glucose less than 55 mg/dL during a period when hypoglycemic symptoms are occurring
- Relief of symptoms with glucose administration

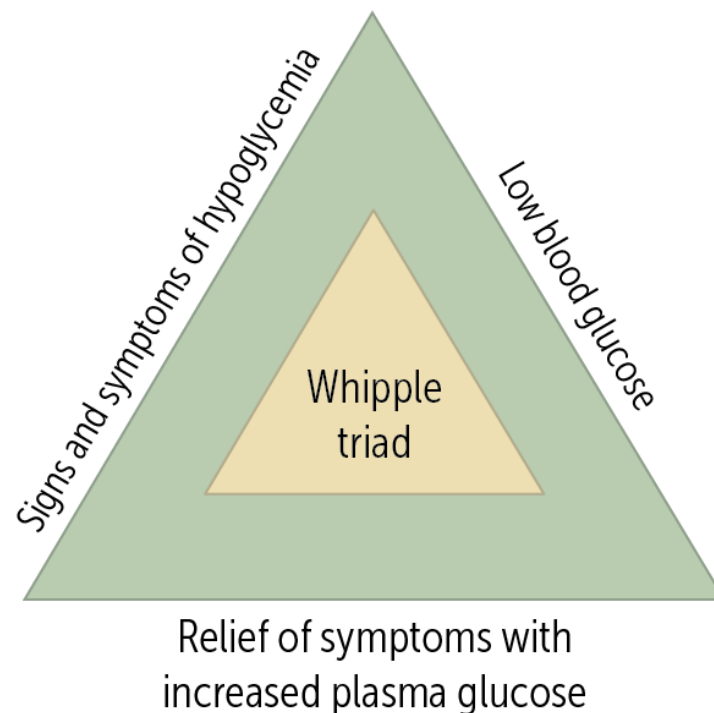


Figure 2. Image Credit: ©ScholarRx, used with permission.

The gold-standard lab test for diagnosing insulinoma is an **elevated insulin level** during either spontaneous or induced hypoglycemia. This is often done using a **72-hour fast**. If the glucose level falls below 50 mg/dL, check both the insulin level and the C-peptide level. Both should be elevated (remember, when insulin is released from vesicles into the bloodstream, it is released along with C peptide).

Imaging studies of the pancreas (CT, endoscopic ultrasound) are also useful in confirming the tumor.

Most insulinomas are surgically resected, whether malignant or benign. Most tumors are benign, so surgery is curative, and the prognosis is good.



Knowledge Check

Question: What three laboratory indices are measured to diagnose insulinoma?

Answer: Plasma glucose, insulin, and C-peptide levels after a 72-hour fasting period are measured to diagnose insulinoma.

What Are VIPomas?

VIPomas are endocrine tumors that secrete too much vasoactive intestinal polypeptide (VIP) into the bloodstream. They arise mostly from non- β islet cells in the pancreas. Even rarer than insulinomas, VIPomas have an estimated incidence of 1 in 10,000,000 people.

Clinical Presentation of VIPomas

The collection of VIPoma clinical symptoms is sometimes referred to as WDHA syndrome, representing watery diarrhea, hypokalemia, and achlorhydria. The diarrhea commonly leads to dehydration. Patients can also present with flushing resulting from the vasodilatory effects of VIP.

Clinical Correlation

WDHA syndrome is also known as pancreatic cholera syndrome because of the similarities between cholera and the symptoms of VIPomas. Both result in large amounts of watery diarrhea, volume loss, and electrolyte abnormalities such as hypokalemia.

Pathophysiology of VIPomas

These tumors secrete VIP without regulation. VIP is responsible for stimulation of small intestine chloride secretion, smooth muscle contractility, inhibition of gastric acid secretion, and blood vessel dilation. Like insulinomas, VIPomas can also be associated with MEN1 syndrome.

Diagnosis and Treatment of VIPomas

Suspect VIPoma in those with unexplained severe high-volume diarrhea, and then check the serum VIP level. A VIP level greater than 75 pg/mL on at least two samples confirms the diagnosis.

Surgical resection is standard therapy, with chemotherapy an option for those with unresectable or metastatic disease.



What Are Glucagonomas?

A glucagonoma is a malignant neuroendocrine tumor of the pancreatic islet α cells that secretes inappropriately large amounts of glucagon into the bloodstream, leading to hyperglycemia. Most are metastatic at the time of diagnosis. The incidence of glucagonoma is estimated to be 1 in 20,000,000 people.

Clinical Presentation of Glucagonomas

Clinical findings in glucagonoma are diverse. Some are related to systemic findings seen in any malignancy (weight loss), whereas others are more specific, such as hyperglycemia and skin lesions.

Knowledge Check

*Question: Glucagonomas present with the **six Ds**: **D**ermatitis (necrolytic migratory erythema), **D**iabetes (hyperglycemia), **D**eep vein thrombosis, **D**eclining weight, **D**epression, and **D**iarrhea.*

Answer: What are the six Ds of glucagonoma?

Hyperglycemia and Diabetes Mellitus. Glucagon causes increased serum glucose by stimulating glycogen breakdown and gluconeogenesis. With unregulated glucagon release, patients develop persistent hyperglycemia and sometimes (40% of these patients) secondary DM. Patients can develop all the secondary complications of DM as well, like retinopathy and chronic kidney disease.

Weight Loss. Weight loss may result from the catabolic effects of high chronic glucagon levels.

Knowledge Check

Question: What causes patients with glucagonomas to lose weight?

Answer: Typically, glucagon leads to glycogen and fat breakdown, leading to weight loss.

Dermatitis. Migratory necrolytic erythema (MNE) is an itchy, painful rash associated with high levels of glucagon. MNE can form plaques and pustules, later with bronze-colored central areas of crusting (Figure 3).



Figure 3. Image Credit: © 2014 Wu et al.; licensee BioMed Central Ltd.

Anyone presenting with this rash who is not being treated with glucagon should be screened for potential glucagonoma.

Venous Thromboembolism. Deep venous thrombosis or pulmonary embolus is present in up to 50% of patients with a glucagonoma. This is likely caused by the increased viscosity of the hyperglycemic blood.

Diarrhea. Diarrhea is seen frequently and may lead to volume depletion.

Psychiatric Symptoms. Depression, psychosis, and agitation are seen in 20% of patients.

Pathophysiology of Glucagonomas

These tumors derive from the pancreatic islet α cells. They secrete large amounts of glucagon, whose main function is to increase the serum glucose level by enhancing glycogenolysis and gluconeogenesis. Metastasis is common, with the liver the most common site.

Diagnosis and Treatment of Glucagonomas

Glucagonomas are diagnosed with the following criteria: persistent hyperglycemia, a plasma glucagon level greater than 500 pg/mL (normal is <50 pg/mL), and any of the other symptoms listed above. CT, octreotide scanning, MRI, and endoscopic ultrasound exams of the pancreas can often localize the tumor.

For treatment, surgical removal of the pancreatic tumor is curative when the tumor is localized and distant metastases are not present. For symptom control, drug analogs of



somatostatin (growth hormone–inhibiting hormone [GHIH]) like octreotide are usually effective in lowering glucagon levels and treating the symptoms (eg, rash and weight loss).

What Are Somatostatinomas?

Somatostatinoma is a tumor of the islet δ cells that results from the overproduction of somatostatin (GHIH). Half of these tumors are found in the pancreas, and the remainder in the duodenum or jejunum. Approximately 75% are malignant. It is the rarest of the tumors we've discussed, with an estimated incidence of 1 in 40,000,000 people.

Clinical Presentation of Somatostatinomas

These tumors are often asymptomatic and picked up on radiologic studies done for another reason. If symptomatic, the most common symptoms of somatostatinoma are abdominal pain and weight loss. Fewer than 10% actually have symptoms related to high somatostatin levels, such as hyperglycemia, DM, or fatty stools (steatorrhea). (Hyperglycemia or DM results from gallbladder disease or the inhibition of insulin release; fatty stools result from cholecystokinin inhibition.)

Pathophysiology of Somatostatinomas

The somatostatin secreted by these tumors acts in a paracrine manner to locally inhibit secretion of many hormones, including insulin, glucagon, gastrin, cholecystokinin, and growth hormone. The inhibition of the gastrointestinal (GI) hormones is most important clinically. Cholelithiasis may result from inhibition of cholecystokinin release, which reduces gallbladder contractility and limits bile entry into the duodenum. Thus, the bile cannot properly digest fatty contents in meals, leading to abdominal pain, bloating, fatty stools, and diarrhea. Digestion is also impaired because pancreatic enzymes and gastrin are both inhibited, leading to low stomach acid.

Diagnosis and Treatment of Somatostatinomas

Suspect somatostatinoma in those with diabetes, diarrhea, and cholelithiasis. In these symptomatic patients, a diagnosis can be made with a fasting plasma somatostatin greater than 30 pg/mL. Often patients have limited or no symptoms, in which case the tumors are detected by imaging as a pancreatic or duodenal mass when evaluating a more common problem like GI bleeding.

Surgery can be curative for localized disease, but, in most cases, the disease has spread by the time of diagnosis. Chemotherapy to control tumor spread and palliative therapy may therefore also be considered.



Clinical Correlation

Interestingly, for unresectable somatostatinomas, the first-line treatment is to use a somatostatin analog like octreotide to inhibit the secretion of somatostatin. These are highly effective at controlling the hypersecretion of hormones, but it is not clear if they act as antitumor agents.

Knowledge Check

Question: How does somatostatin physiologically regulate most GI hormones?

Answer: Somatostatin physiologically regulates most GI hormones by inhibiting them.

Case Connection

Thinking back to SG, what do you tell her about the first step in her treatment? “The short answer is yes; it can be removed surgically,” you say. “But a complete treatment is more complicated than this.” You explain that her glucagonoma is part of a multiple endocrine syndrome that is also causing high serum calcium due to high parathyroid hormone levels (explaining the constipation). You then say that you want to do more testing to determine whether the tumor has spread to other parts of her body. “In the meantime, you can take medications called somatostatin analogs to help your symptoms.” You give SG time to ask questions and refer her to an endocrine surgeon.

Summary

- Endocrine tumors arise in the islet cells of the pancreas, whereas exocrine tumors arise in the exocrine portion of the pancreas.
- Endocrine tumors can be functional (75% of tumors) or nonfunctional (25%); functional tumors are named after the hormones they produce.
- Insulinomas secrete insulin and arise from the β cells of the pancreas.
- Insulinomas typically present with hypoglycemia (confusion, headache, disorientation, coma).
- VIPomas arise from the non- β islet cells of the pancreas and secrete vasoactive intestinal polypeptide.
- VIPomas can cause watery diarrhea, hypokalemia, and achlorhydria (WDHA syndrome).



- Glucagonomas arise from the α cells of the pancreas and secrete glucagon.
- Glucagonomas present with depression, weight loss, glucose intolerance, deep vein thrombosis, and necrotizing migratory erythema.
- Somatostatinomas arise from the δ cells of the pancreas and present with diabetes, gallbladder disease, diarrhea, and steatorrhea.
- Surgical resection of somatostatinomas is first-line treatment and can be curative; for those with metastatic disease, however, the prognosis is poor.

Pancreatic Tumors - Review Question

A 3-year-old boy was recently diagnosed with what is suspected to be a VIPoma on imaging. In preparation for his admission to the hospital, the clinician reviews the complications of VIPomas. Which of the following combinations of abnormalities is seen in a patient with a VIPoma?

- Diabetes, cholelithiasis, and diarrhea
- Hypoglycemia, disorientation, and coma
- Watery diarrhea, hypocalcemia, and achalasia
- Watery diarrhea, hypokalemia, and achlorhydria
- Weight loss, deep vein thrombosis, and dermatitis



Diabetes & Other Disorders of the Endocrine Pancreas- Review Questions and Answers

Diabetes Mellitus: Foundations and Frameworks - Review Question

Which of the following tests provides the most accurate measurement of glycemic control over time?

- Fasting glucose
- Finger-stick glucose
- Glucose on urine dipstick
- **Hemoglobin A_{1c} (Correct Answer)**
- Oral glucose tolerance test

Feedback: The correct answer is hemoglobin A1c. HbA1c provides a long-term measurement (a 3-month period) of glycemic control, and the level is linearly related to the fasting glucose level, averaged over time. A single fasting glucose level can be a helpful measurement for short-term glucose control, but it does not provide any information about the long-term trend of glycemic control. Likewise, a single finger-stick glucose is also a snapshot in time of what someone's glucose level is, but it does not provide any long-term information. A glucose on urine dipstick is a test that measures glucose levels in the urine but is not useful for long-term monitoring, because glucose only appears in the urine when the blood level exceeds 240 mg/dL, which is above the "good-control" threshold. An oral glucose tolerance test is occasionally used for initial diagnosis but is cumbersome and not used to follow up with patients who are receiving treatment.



Type 1 Diabetes Mellitus - Review Question

Which of the following clinical findings is most likely to be present in a patient presenting with diabetic ketoacidosis?

- Blood pH of 7.55
- Coma
- **Deep and rapid breaths (Correct Answer)**
- Normal to elevated insulin levels
- Seizures

Feedback: The correct answer is deep and rapid breaths. These are commonly called Kussmaul respirations, which are the body's way of compensating for metabolic acidosis. The blood pH in ketoacidosis would be <7.3 , not >7.55 . Coma is an uncommon presentation of DKA and is more often seen in hypoglycemia. Normal to elevated insulin levels are found in patients with insulin resistance who have type 2 DM, not in DKA. Seizures are uncommon in DKA.

Type 2 Diabetes Mellitus - Review Question

Which test results would best confirm a diagnosis of type 2 diabetes mellitus in a symptomatic patient?

- Blood hemoglobin A_{1c} result of 4.1%
- Overnight monitoring of glucose levels at an accredited center
- **Random blood glucose test showing a value of >200 mg/dL (Correct Answer)**
- Serial glucose blood tests for a week
- Two fasting blood glucose test results >110 mg/dL

Feedback: The correct answer is a random blood glucose test showing a value of >200 mg/dL. A blood hemoglobin A_{1c} result of 4.1% would suggest a person who is healthy and not at risk for diabetes mellitus (DM). Overnight monitoring of glucose levels at an accredited center and serial blood glucose tests for a week would be excessive in trying to establish a diagnosis. Two fasting blood glucose test results >110 mg/dL would suggest prediabetes, not overt DM.



Diabetic Ketoacidosis - Review Question

Which of the following is most likely to occur after the administration of insulin therapy in a patient with diabetic ketoacidosis?

- Magnesium moves out of cells
- Phosphate moves out of cells
- **Potassium moves into cells (Correct Answer)**
- Potassium moves out of cells
- Sodium moves into cells

Feedback: The correct answer is potassium moves into cells. Insulin treatment results in potassium moving into cells because insulin stimulates Na^+/K^+ -ATPase, which transports potassium from the blood into the cell. Magnesium moves out of cells is incorrect, as magnesium does not move out of cells to a substantial degree from insulin therapy, and magnesium itself sometimes needs to be supplemented. Phosphate moves out of cells is likewise incorrect. Phosphate is like potassium in that insulin permits its entry into cells. Potassium moves out of cells is incorrect because potassium does not move out of cells from insulin therapy. Sodium moves into cells is incorrect, as sodium does not move into cells to a substantial degree from insulin therapy.

Part Hyperosmolar Hyperglycemic State - Review Question

A patient has poorly controlled type 2 diabetes mellitus and presents to the emergency department with polyuria, polydipsia, blurred vision, and a serum glucose concentration of 780 mg/dL. Which of the following findings is most likely to be found on further evaluation?

- **Decreased skin turgor (Correct Answer)**
- Deep (Kussmaul) respirations
- Elevated jugular venous pressure
- Fruity breath odor
- Low blood pH

Feedback: The correct answer is decreased skin turgor. Hyperosmolar hyperglycemic state (HHS) and diabetic ketoacidosis (DKA) are characterized by volume depletion caused by osmotic diuresis, causing decreased skin turgor and decreased jugular venous pressure (JVP). Deep (Kussmaul) respirations are seen in DKA as a hyperventilatory response to acidosis but would not be found in HHS. Elevated jugular venous pressure is not correct because the decreased volume status of the patient would lead to decreased JVP. Fruity breath odor is found in DKA because of exhaled acetone but is absent in HHS. Low blood pH is seen in DKA but is less often seen in HHS.



Chronic Complications - Review Question

Which is the most common microvascular complication of diabetes mellitus?

- Coronary artery disease
- Diabetic ketoacidosis
- Diabetic kidney disease
- Peripheral neuropathy
- **Retinopathy (Correct Answer)**

Feedback: The correct answer is retinopathy. Statistically, retinopathy is the most common microvascular complication of DM. Coronary artery disease is macrovascular, and diabetic ketoacidosis is an acute complication. Although diabetic kidney disease and peripheral neuropathy are microvascular complications of DM, they are less common than retinopathy.

Hypoglycemia - Review Question

Which of the following would most likely cause hypoglycemia?

- **Decreased gluconeogenesis (Correct Answer)**
- Decreased insulin
- Glucagon-secreting tumors
- Increased cortisol
- Increased glycogenolysis

Feedback: The correct answer is decreased gluconeogenesis. Serum glucose levels are increased by the formation of glucose from substrates such as amino acids and lactate, a process termed gluconeogenesis. Serum glucose levels are also increased when stored glycogen is metabolized to glucose (glycogenolysis). These processes are inhibited by insulin and stimulated by hormones such as cortisol, epinephrine, and thyroid hormone. Decreased insulin, glucagon-secreting tumors, increased cortisol, and increased glycogenolysis would all increase blood glucose levels.



Pancreatic Tumor – Review Question

A 3-year-old boy was recently diagnosed with what is suspected to be a VIPoma on imaging. In preparation for his admission to the hospital, the clinician reviews the complications of VIPomas. Which of the following combinations of abnormalities is seen in a patient with a VIPoma?

- Diabetes, cholelithiasis, and diarrhea
- Hypoglycemia, disorientation, and coma
- Watery diarrhea, hypocalcemia, and achalasia
- Watery diarrhea, hypokalemia, and achlorhydria
- Weight loss, deep vein thrombosis, and dermatitis

Feedback: The correct answer is watery diarrhea, hypokalemia, and achlorhydria. VIPoma causes WDHA syndrome, which includes these abnormalities, due to the excessively watery diarrhea and potassium and bicarbonate wasting. Diabetes, cholelithiasis, and diarrhea can be seen in somatostatinomas. Hypoglycemia, disorientation, and coma can be seen in insulinomas. Watery diarrhea, hypocalcemia, and achalasia is not the combination of abnormalities seen in VIPomas; specifically, hypocalcemia and achalasia—an esophageal disorder—are not seen. Weight loss, deep vein thrombosis, and dermatitis can be seen in glucagonomas.